

'This is an outstanding collection of essays which invites a radical rethinking of photography. Each chapter dismantles conventional understandings of photography by examining in detail a specific assemblage of social practice, camera technology and light-generated image. What photography is, what it does and what it might do is thus rendered radically open, and photography is once more made as remarkable, emergent and diverse as it was a century and a half ago. Essential reading for anyone interested in photography and visual culture.'

Gillian Rose, *Professor of Cultural Geography, The Open University*

'This exciting and multifaceted book casts new light on the practice of photography. Highlighting the various processes of communication, networking and human-nonhuman relationality in different parts of the world, it shows the photographic medium as literally teeming with life. This is a must-read not just for scholars and students of photography but for anyone who reads the news, uses social media, moves from place to place or owns a camera phone!'

Joanna Zylinska, *Professor of New Media and Communications at Goldsmiths, University of London*

Digital Photography and Everyday Life: Empirical studies on material visual practices explores the role that digital photography plays within everyday life. With contributors from ten different countries and backgrounds in a range of academic disciplines – including anthropology, media studies and visual culture – this collection takes a uniquely broad perspective on photography by situating the image-making process in wider discussions on the materiality and visibility of photographic practices and explores these through empirical case studies.

By focusing on material visual practices, the book presents a comprehensive overview of some of the main challenges digital photography is bringing to everyday life. It explores how the digitization of photography has a wide-reaching impact on the use of the medium, as well as on the kinds of images that can be produced and the ways in which camera technology is developed. The exploration goes beyond mere images to think about cameras, mediators and technologies as key elements in the development of visual digital cultures.

Digital Photography and Everyday Life will be of great interest to students and scholars of Photography, Contemporary Art, Visual Culture and Media Studies, as well as those studying Communication, Cultural Anthropology, and Science and Technology Studies.

Edgar Gómez Cruz is a Vice-Chancellor Research Fellow at RMIT, Melbourne.

Asko Lehmuskallio is Chair of the ECREA TWG Visual Cultures and founding member of the Nordic Network for Digital Visuality.

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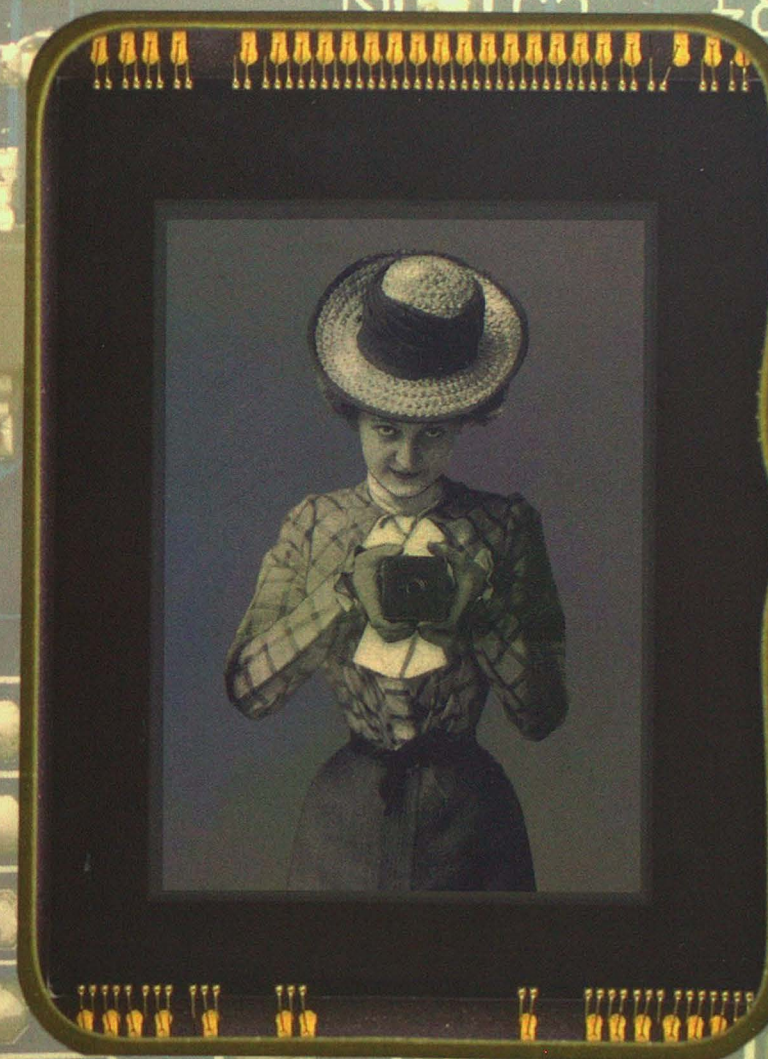
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Digital Photography and Everyday Life

ROUTLEDGE

Empirical studies on material visual practices



Edited by **Edgar Gómez Cruz**
and **Asko Lehmuskallio**



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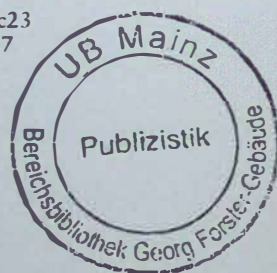
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Foreword

Richard Chalfen

We live in an intellectually challenging time for witnessing, participating in and studying the emergence of new media, especially digital media immediately connected to problematic issues of visibility. Problematic concern lies within contexts of visual culture, human interaction, social relationships and competent participation in everyday life. Digital capabilities continue to contribute to camera-mediated sociality in different and more varied ways than initially expected. The pace of technological change has been impressive and begs the question: Have interests and abilities of the scholarly community kept pace in theoretical and practical ways?

Any review of the historical development of a research field suggests significant questions and faces important challenges when the subject matter of such a field is rapidly developing and changing, when definitions of such subject matter are either missing or being altered along the way and, third, when recognition and acceptance of academic attention to such subject matter are also in flux. In short, several dimensions or storylines must be included and simultaneously interwoven when constructing a dynamic assessment of such a field.

Fundamental questions

What consistencies and changes can be noted? First and foremost lies a neglected problem: what is the field being discussed and does this field have specific origins, themes, definitions, parameters and the like? Can we claim that this field is conceptually, methodologically, materially defined and unified? Or, is it possible to feel there is enough of a shared understanding about a hypothetical field that participants are able to contribute? Most importantly, what have researchers wanted to learn and know? The general answer for this last question is: *What do ordinary people do with their cameras and personal pictures, as part of everyday life?*

Nomenclature ambiguity

'What exactly is being studied?' Several commentators have cited instances of terminological confusion and shortcomings with nomenclature and the ambiguous use of certain labels. These include such moving targets as 'everyday', 'family', 'amateur', 'private' and 'public', 'home media', and 'vernacular', while some deride 'artistic' as the primary reference point for comparisons. Finally, as picture-taking changes from analogue to digital, some have even challenged the appropriate use of the term 'photographer'; the legitimacy of 'photography' and 'photographic' has been questioned, given the lack of chemically based processes and, as emphasized in the following chapters, a more expansive configuration of technologies, affordances/constraints, institutions and practices.

Multiple perspectives

The emergence of a shared and evolving understanding of everyday photography owes much to a growing diversity of academic perspectives. Our immediate attention is drawn to what is now recognized as the visual social sciences, including early studies in sociology, communication and anthropology, as well as folklore. Other contributions have been made by psychologists, historians, cultural geographers, family therapists and the like. Courses on personal amateur photography have emerged from such disciplines as English, Communication, Folklore, American Studies, History and Material Culture Studies, among others. A lot of attention is found in Information Science and Computer-Human Interaction (CHI), resulting in an increasingly rich and relevant bibliography of scholarly articles and references, further enhancing a field of study. History, for example, specifically the history of photography, has enlarged its scope to include multiple histories of photographic expressive forms, encouraging discussions of vernacular forms of snapshot activities and delving further into the emergence of everyday, at-home photographic practices. Interestingly, the humanities have played roles – even novelists, poets and song writers have contributed observations on the significance of everyday photography. In short, interests within different academic orientations have produced a broader range of questions, methodologies and insights.

One trend has been the slow acceptance of the scholarly study of everyday/non-professional photography. Early on, studies in home media were not a part of the canon and generally not an appropriate topic of academic inquiry. Perhaps the material was considered too trivial, too common and all too embedded in everyday life and, hence, taken for granted. However, much has changed since the first home media studies; the enthusiastic acceptance of digital technology, for everyday photography, is dramatically reflected in changing research questions and scholarship in general.

Approaching empirical work

First, we should acknowledge the interests of film and camera companies, their marketing firms and advertisers, who demanded empirical data prone to statistical results and analyses. The second source of potential influence is provided by the popular photography magazine industry, for example, the pages of *Popular Photography* and *U.S. Camera*, offering advice for making better pictures than family snapshots and avoiding the 'snapshot look' as much as possible. However, these sources seldom address the kinds of questions most relevant to social scientists and the problematic ways of working with these pictorial materials as sources of data.

More relevant is a growing range of research perspectives including paradigms and methods of study. Uneven attention was given to historical, philosophical, psychological, social, cultural or purely personal circumstances, with minimal attention to systematic empirical observation. In turn, we also find very uneven paths of academic attention to what ordinary people did, or currently do, with their cameras and pictures. Application of different approaches has been varied, for example, ethnographic description, content analysis, semiotic analysis, participant observation, discourse analysis, feminist approaches, etc. and spread across a growing number of disciplines and interests, all with varying value placed on including verbatim comments by family members and a greater respect for empirical approaches to field data.

Early empirical studies are rare, leaving much discussion of everyday photography in the form of personal reflections, mediations, narratives and explorations of such general questions as: *What do photographs mean? How do we understand the story of our lives through the pages of the family photo album? How do personal pictures exert a power to shape our personal memories and self-conceptions?* Less fieldwork-oriented observers chose to offer reflections on their own personal photographs, produced by the observer's family. Some accounts consisted of personal meanings of just a few images; initial intuitions and speculations were insightful and occasionally led to theoretical and philosophical discussion. The work of Roland Barthes and Susan Sontag provide two well recognized and frequently cited examples. Some accounts have used a sample-of-one (e.g. Barthes' 'Winter Garden Photograph') when an author preferred to reference his/her own picture collection as a 'spectator', speculating that everyone had the same or very similar pictures, feelings, understandings, and then went on to generalize from there. There was a reluctance to do fieldwork on everyday photography, to collect data or to apply replicable methods of examination, description and analysis to a collection of material.

An important exception is the empirical work done in France by Pierre Bourdieu. His field research included a blend of surveys, interviews and ethnographic data, and set the stage for asking what people do *with* their

cameras and what pictures do *for* people who take and keep them. He attended to picture-taking habits and social functions of family photography in the aid of understanding social integration, social class and the French stratification system.

Scholarly attention to taking personal photographs seriously grew from the 1970s to recent decades, including the work of Julia Hirsch, Jo Spence, Gillian Rose, Marianne Hirsch, among others, all working with varying degrees of empiricism with analogue photography. Richard Chalfen offered a rare case of integrating questionnaires into his fieldwork. Early empirical work tended to focus on understanding the content of people's pictures and applying functional explanations for why people took these photographs and what such collections did for current and future family members. There was much less attention paid to the process underlying the production, use and display of such material. Early academic interests were much narrower than we see today, as abundantly illustrated in the chapters comprising this book. Historically, shifts in scholarly attention reveal that studies of what early family-related pictures *looked like* moved into questions about how ordinary people *enacted/practised* family photography – what family members did with their cameras and what their camera use did for family life, for social cohesion, kinds of interpersonal and cross-generational communication and the like.

Roles of cameras in everyday life

Other views emerge from considering changes in the roles that cameras and pictures played and how cameras became embedded in everyday life and, in turn, how digital cameras might be changing the home culture of ordinary people. Most would easily agree that dramatic changes have occurred in quantitative dimensions of camera possession, picture-taking frequency and the electronic display of personal photographs. Historically, examples of home media have been increasing in frequency and, in turn, steadily 'leaving home' as we find private pictures appearing in public zones of appreciation; facilitation of picture sharing has drastically increased the density of connectivity accompanied by a decline in the volume of printed personal pictures.

Everyone eagerly points out that digitization has clearly increased the number of cameras (and, of course, phones) per family unit. Even before camera phones, marketing reports indicated that numbers of both analogue and digital cameras, as well as frequency of picture-taking per household, had been growing. Digitization appears to have caused photography to be less compartmentalized in everyday life. There is no one 'family camera', primarily operated by Dad, just as no one person is assigned the role of family photographer. When the time comes to take a trip, family members are less likely to ask *Did you remember the camera?* or *Anyone buy some film?* Having cameras in the hands of all family members

sometimes inspires the problematic and frequently cited question *When should I give my kid a cellphone?*

Second, camera appearance and use are no longer restricted to special occasions. In contrast, they are virtually carried everywhere, for potential use in 'non-traditional', newly 'photogenic' settings, events and activities. Any scene can be designated as 'photo-worthy' and will receive multiple recordings, multiple takes, from many people. Cameras are being called upon to see and 'capture' more of the world, of everyday life, than ever before. Indeed, more settings and topics are producing many instances of 'sousveillance' which, on occasion, are used as citizen photojournalism. Third, digital pictures seem to be more frequently accessed and referenced in everyday conversations, with fewer calls for 'getting out the family album'.

Perhaps the most dramatic change centres on a range of activities suggested by the younger members of families. Importantly, some scholars have refocused their attention from the pro-social norms of family-based adults and parents to occasionally anti-social habits produced by community-based younger people. Inspired by boundary-testing characteristics of adolescence, their ways of combining use of camera phones and the internet have, for instance, produced troublesome 'sharing' of photographs in instances of sexting and cyber-bullying.

Redirections for academic attention

My sense is that empirically based scholarship has been slow to keep pace with the rapid pace of emerging camera technology and, second, creative uses of such apparatus within everyday life. We are prompted to ask how changes associated with digitization require or result in altering scholarly directions or changes in research strategies, including methods of observation, inductive or deductive reasoning, quantitative and qualitative methodologies and the like. Just as digitization prompted the entrance of non-camera companies into photography, we also saw the appearance of new interests from different academic disciplines.

One theme that spans past, recent and hopefully future studies is how everyday photography functions as a mode of visual communication, one significantly embedded in visual culture. Some maintain a difference between the sporadic making of memories for long-term family uses (connected to analogue photography) and the more frequent recording and sharing of experiences for short-term individual uses (characteristic of much contemporary digital photography). Significantly, both instances involve participating in acts of visual communication, whether for interests of participants 'in the moment' or for shared memories of family members across generations.

Scholars will sense a need for new models of pictorial/visual communication for understanding the cultural underpinnings of what's happening,

conceptually and practically, in terms of product, process and/or practice. Cross-cultural studies of home media can serve to avoid ethnocentric assumptions that prematurely homogenize constituencies, in this case the expanding universe of ordinary people acquiring a camera in their new mobile phone and taking increasing numbers of pictures as part of their everyday lives. Participant observation and ethnography will play important roles in such efforts, as nicely discussed in several of the following chapters.

One fear is that increased attention to schemes and patterns of affordances takes us away from fieldwork and empirical studies that reveal what people actually know about what *can* be done with their new equipment, in conjunction with or in contrast to what these same people are actually doing with such opportunities, to stretch their practices of personal photography in everyday life. In short, interest should persist on what ordinary people, both adult and especially young people, are experiencing with their new 'electronic possibilities', what they might know about algorithmic 'interference' and, in turn, what such changes might be doing to them. Clearly, a blend of these interests is apparent in the following collection of chapters.

This collection of chapters offers a productive approach along several dimensions, principally: (1) to multiple, often cross-cultural, applications of smartphones as just one example of modern camera technology, and (2) to new meanings of 'photography' based on enhanced understandings of process and procedures, in relation to the varied and multiple results of an imaging process or, as Edgar Gómez Cruz notes in his chapter, 'vernacular photography practices are increasingly becoming algorithmic and a source of metadata, while expanding their function as depictions and representations'. Editors Lehmuskallio and Gómez Cruz have presented us with a broad overview of some of the main transformations digital photography is bringing to everyday life, coupled with ways of productively thinking about and studying complex ongoing changes in an alignment of technologies, meanings, uses, institutions and practices, all central to evolving digital culture. As I initially stated, it is a very exciting time to be part of visual digital culture.

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7 Photographs of place in phonespace

Camera phones as a location-aware mobile technology

Mikko Villi

Introduction

Verbal mobile phone communication is mostly confined to locations that have no intrinsic relationship to the act of communication; the content of communication is mostly determined by the participating subjects, not by the physical location in which the communicators reside. When talking or texting on the mobile phone, one does not necessarily have any indication of the location of the other communicator. By contrast, when communicating with photographs, there is often an essential connection between the physical setting and the act of communication. The photograph is, after all, captured in a definite location and, with the aid of the mobile phone, the photograph can be shared right from the spot immediately after capture. By using a camera phone photograph, one can 'send the place', visually mediating one's local presence. The main argument of this chapter is that camera phone photographs accentuate the significance of location in mobile communication, from being mere points of communication into being individual places, places with identity.

In this sense, the sharing of camera phone photographs is a predecessor to, for example, such mobile applications as Swarm¹, which enable the easy sharing of location and also emphasize the significance of the location where the use of the mobile phone occurs. An important contribution of the chapter is, then, in shedding light on such location-based networked interactions in mobile communication that existed before smartphones with global positioning system (GPS) receivers were introduced to consumers. A goal of the chapter is also to build a theoretical link between camera phone photo communication practices and novel location-aware mobile technologies. *Location-aware technologies* consist of devices that are able to locate themselves via a GPS, Wi-Fi or the triangulation of radio waves, and therefore can provide users with location-specific information (de Souza e Silva & Frith 2012, p. 6). *Locative media* can be defined as a term that is used to capture the diverse array of location-aware technologies and practices (Wilken & Goggin 2015, p. 4). Importantly, as already noted, photography is also a strongly location-aware

form of communication, although it has not generally been studied in relation to locative media.

The concepts of place and location often seem to overlap in discourse on both location-aware mobile technologies and photography. Also, in everyday language, the two words are commonly used rather synonymously. Adriana de Souza e Silva (2013, p. 119), notes how the term *location* has received sparse scholarly attention when compared to its correlate *place*. For the purposes of this chapter, it is necessary to distinguish the two and offer a specific definition of place (see also Pink in this volume). First of all, location can be considered to be a place deprived of meaning (Cresswell 2004). Location refers merely to the position of a person or thing and is, thus, less culturally and socially encompassing than place (Wilken & Goggin 2015, pp. 2–3), which is all-pervasive in the way that it informs and shapes everyday lived experience (Wilken & Goggin 2012, p. 6). *Place* does not refer only to a point on a map but rather arises in the dynamic interrelatedness of things (Malpas 2012, p. 28) and is, thus, about more than just physical geography (Hjorth 2012, p. 140). Not all places even have fixed geographical coordinates. For example, heaven and hell are places but cannot be located (de Souza e Silva & Frith 2012, p. 10).

Place is a networked, dynamic and meaningful entity and a psycho-social construct that includes social relations (Gordon & de Souza e Silva 2012, p. 90; Gibson, Luckman & Brennan-Horley 2012, p. 123). In a sense, the relation between location and place can be outlined by comparing location to the concept of ‘non-place’, as coined by Augé (1995). Certain locations, such as motorways, airports and aeroplanes, hotel chains, leisure parks and shopping malls are physical places but at the same time they are non-places. Augé (pp. 77–79) explains that a place which cannot be defined as relational, historical or concerned with identity is a non-place.

A useful concept in examining the mediation of place in both mobile communication and photography is ‘mediated presence’ (Villi 2015). Mediated presence is not a clearly articulated theoretical model, but rather a conceptual theme describing the use of telecommunication technology for being in contact with one another over physical distance – being socially present despite being physically absent. A photograph, too, can mediate presence, as it embodies the possibility of the object, place or person in the photograph being present for the viewer. In the context of camera phone communication, these two dimensions of presence are conjoined, as presence is mediated by images over distances by utilizing a tele-communicative connection.

Re-localization of mobile communication

The popularity of *location-based services* – services and media of communication that are functionally bound to a location (Wilken 2012, p. 243) – is an interesting trend in mobile communication, as it is in striking contrast

to a central theme in the discourse on mobile communication, namely the liberation from location in communication. In telephone communication, mobility is nowadays more of a premise than an added value. A popular notion has been that, when people become immersed in the 'mobile media sound bubbles of communication' (Bull 2005, p. 178), the places they pass through in their daily lives increasingly lose significance (Meyrowitz 2005, pp. 25–27; Geser 2004, p. 36; Ling 2008). It is true that the mobile phone does have the capacity to disrupt a user's connection to place and co-present others. However, at the same time, it can connect the user to (even previously unknown) others in that place and to the place itself (Campbell 2013, p. 11).

In contrast to many of the views presented in the first decade of the century, mobile phone use is now increasingly bound to a location. Rather than directing someone towards 'absent presence' – as in indicating a state where one is physically present, yet at the same time absorbed by a technologically mediated world elsewhere (Gergen 2002, p. 227) – mobile technologies actually work as interfaces to locations. The technologies also foreground the importance of locations – as places – in our social and networked interactions. Therefore, due to the increasing use of location-aware technologies, location becomes paramount to social interaction and spatial construction, and locations are filled with a wealth of digital information (de Souza e Silva & Frith 2012, pp. 8, 14, 102, 169).

Mobile devices can help strengthen users' connections to their surrounding location, rather than removing them from it. Location-based services have become an elemental component of mobile communication. By using the mobile phone's location-aware capabilities, users can annotate locations, find other people in the vicinity and access information connected to specific locations (de Souza e Silva 2013, p. 117; Humphreys 2013). When people attach visual or other information to locations they share their memories and understandings of those locations, along with the physical coordinates (Özkul 2015, p. 102). According to what Gordon and de Souza e Silva (2012) call 'net locality', location is becoming the organizing logic of our networked interactions. Net localities include an increasing amount of location-based information that is not only attached to places but also becomes an intrinsic, dynamic part of those places. Through the increasing use of location-aware technologies, locations acquire renewed interest and meaning (de Souza e Silva & Frith 2012, p. 10). 'De-localized' mobile communication is 're-localized' (Licoppe & Heurtin 2002, p. 101), either automatically or as a consequence of the intentional activities of the user. This re-localization represents a new awareness of location in mediated communication.

The re-localization of mobile communication can be accomplished in various ways. In a verbal call, re-localization can be achieved by providing information on the caller's location during the phone call. 'Location talk' has been an obvious feature of mobile communication. For this reason,

much research into mobile technologies has addressed issues focusing on location and place (Laursen & Szymanski 2013, p. 315). Nowadays, smartphone apps offer a multitude of localization services, ranging from Swarm to Find My Friends. The users of Swarm can even 'collect places' when checking in takes precedence over movement (Gazzard 2011, p. 416).

Camera phone users, for their part, are localized by the photographs they communicate. In its most simple form, the physical location is appended by automatic geotagging of the photograph by embedding meta-data received from the phone's GPS chip. In mobile phone photography and such smartphone apps as Instagram, geotagging is increasingly the default, rather than a choice (Hjorth & Pink 2014, p. 40). Naturally, photographs also communicate the location by the contents of the photograph itself.

Mediated presence

Photographic re-localization is closely connected to the concept of mediated presence. In general, mediated presence is enabled by the various tele-technologies – television, radio, telephone and the internet. Mediated presence is not about direct physical co-presence but, instead, about presence that is communicated through a medium. Users of mobile communication technology are not present by any physical means, but merely connected to one another in another location by using a communication device. Mediated presence emulates the experience of actually being in a remote location or with a remote other.

Licoppe (2004, p. 135) states that telecommunication technologies are used in mediated relationships to substitute or compensate for the rarity of face-to-face interactions and the absence of others. Therefore, the aspect of *tele* in communication can actually be equated with absence. 'The essence of tele-presence is that it is anti-presence' (Manovich 1995). Mobile phone communication is about technologically mediated sociability (Schroeder 2010, p. 83; Jensen 2013, p. 26; Ito & Okabe 2005, p. 263) in times of absence (Haunstrup Christensen 2008, pp. 435, 446). Mediated presence, thus, captures the most salient trait of mobile communication: the possibility for close communicative engagements among distant agents, a form of absence in which proximity can be established and preserved through mediated communication. The fundamental ambivalence of telecommunication media consists of their usability as tools for both presence and absence: they can induce a feeling of presence and facilitate absence (Villi & Stocchetti 2011, pp. 102, 104).

Similarly, a photograph conveys the presence of the absent one, the object or person captured in the photograph. Photography exemplifies the epistemological dialectic of presence and absence (Lister 2007, p. 353). Actually, as Sonesson (1989, p. 73) writes, 'The whole point of photography is to offer us vicarious perceptual experience, that is, the illusion of

having seen something without having been present at the scene.' A photograph gives presence to absence. Sontag (1977, pp. 16, 155) asserts that a photograph is both a 'pseudo-presence' and a 'token of absence', a surrogate possession of a cherished person or thing.

In the presence mediated by a photograph, of great importance is the indexical relation between the photograph and the object or location photographed. The index ties the photograph to the location. Barthes ([1981]/2000, pp. 76–77, 80, 87) notes that every photograph is a certificate of presence, 'literally an emanation of the referent'. The fact that the photographed object has been there is the essence, the 'noeme' of photography. As its name translates from Greek, the photograph has been 'written with light' (Batchen 1999, p. 101). Photographs are 'images produced as a consequence of being directly affected by the objects to which they refer' (Batchen 2004, p. 31). A photograph is not only an interpretation of the real but also a trace, something directly stencilled off the real. There is always a presumption that something exists in the location, or did exist, which then is depicted in the picture (Sontag 1977, pp. 5–6, 154).

What the mobile phone adds to the locally rooted, indexical presence is the contemporaneous aspect of communication. The mobile photograph, sent and received in a matter of minutes, is not rooted to the location where the photographer was, say, two weeks ago, but where he or she is at the moment of communication. The novel element of telecommunication affects the way a photograph can mediate the being of the object or person in the image. A conventional photograph mediates it from *there-then* to the *here-now* (Barthes [1977]/1991, p. 44), as for example, when a photographic print is here in front of my eyes (present), but at the same time the situation depicted in the photograph is already gone (in time) – the person photographed might even be dead. By contrast, a photograph sent forward from a camera phone immediately after capture can form a connection between *there-now* and the *here-now* (Villi 2015).

A dividing line can be drawn between mediated presence as the sense or the illusion of 'being there' and 'being there together' – between 'physical presence' and 'social presence' (IJsselsteijn *et al.* 2000; Schroeder 2005, p. 342). For the major part, mobile communication can be described, essentially, as using a medium to be with another, providing a social presence. More often than having a sense of the place, the users experience the sense of being with another (Biocca, Harms & Burgoon 2003, pp. 456, 458). However, with the mobile photograph, it is possible also conveniently to send the place and give the other a sense of being there. The photograph does not necessarily need to express photographic skills or quality, as its function is more in the communicative use of the photograph. The 'visual little nothings' of everyday life, photographs of and from mundane places, serve well in establishing a connection and communicating one's presence. Sending the place forms a part of 'visual chit-chat' (Villi 2012), as people communicate about themselves and their

whereabouts with a steady stream of photographs, such as is exemplified by the selfie culture. Visual chitchat or small-talk can be a valuable communicative activity, even if it consists of repetitive and superfluous photographs. Photographs enable phatic interpersonal communication about place (Sutherland 2012, p. 165). These conventions follow largely from the practices of mobile communication, where often the act of calling or texting counts at least as much, if not more, than what is said.

'Phonespace' and 'photospace'

In relation to place and location, it is essential to discuss space as well. Here, I utilize *space* in the sense that Massey (2008) defines it: space is the product of the simultaneous existence and interrelations of people. I adapt this notion of space to 'telecommunicative space', where the interrelations do not occur between co-present people, but rather between people connected via telecommunication media. The concepts of telecommunicative space and social presence describe very much the same thing. Thus, people reside in a physical location, but at the same time they can act in a telecommunicative space, for example when talking on the telephone. Castells *et al.* (2007, pp. 171–178) point out how mobile communication redefines places as a space of communication; places act as points of convergence in communication networks and they are subsumed into the space of flows.

The space of the telecommunicative connection can also be labelled as 'phonespace' (Hulme & Truch 2005, p. 466). People spend an increasing share of their daily lives in the phonespace. A growing proportion of social interactions are mediated; they occur with mostly verbal representations of others, rather than through co-present face-to-face encounters (Ijsselsteijn, van Baren & van Lanen 2003). For many users, mobile communication can be perpetual rather than consisting of only temporal excursions into the telecommunicative space. Ito and Okabe (2005, p. 264) conclude that by using mobile communication, people can experience a sense of a persistent social space. Thus, the phonespace is a constant space, but the physical locale keeps on changing. With increasing mobile internet connections, phone communicators move between online and offline, surrounded by people and things, while also traversing social networking sites and other online elements. For them, places are, in fact, both online and offline places through these movements (Hjorth & Pink 2014, p. 45). Mobile media mesh located place and networked space (Richardson & Wilken 2012, p. 184).

Like location, the telecommunicative space, as such, has no identity. It obtains its value and importance from the fleeting and ephemeral, yet meaningful, acts of communication that take place (or perhaps I should say 'take space') in the communicative space. Many 'real places' (e.g. traditional meeting places such as cafés) – places of active presence – can actually turn into non-places when people engage in mobile communication with those who are absent and thereby transform their physical locales into

socially meaningless places. Mobile communication can transform non-places into telecommunicative spaces and places into non-places. However, through the use of both location-based services and photographs in mobile communication, both places and non-places can acquire new location-related meanings. First, many places become locations – that is, the locational aspects of many places acquire relevance (de Souza e Silva & Frith 2012, p. 10). Second, many non-places, or the ‘in-between places’, are no longer contexts for just killing time. The temporal presences in these non-places have become key moments where new forms of visibility and sociality are generated through camera phone photography and the digital co-presence enabled by mobile media (Hjorth & Pink 2014, pp. 42, 45). For example, on Swarm, users are deliberately marking locations, such as airports and railway stations, as places of identity by sharing them with others. This identity is then extended through the social network attached to Swarm (Gazzard 2011, p. 408).

Sending the place

In the following, I will present observations and perceptions from a study that I carried out with a group of Finnish camera phone users. For the study, I utilized qualitative in-depth interviews, interviewing eight subjects individually. The sampling procedure for the interviewees was purposive and consisted of searching for exemplary informants – people who have shared photographs from their mobile phones. The interviews lasted, on average, one-and-a-half hours, and they were recorded and transcribed.

I used a semi-structured model for the interviews and presented a set of questions to all of the interviewees. The dialogue during the interviews was staged according to a thematic, topic-centred structure, which also provided a context for unexpected themes to develop (Mason 2002, pp. 62–63). The interviewees could continue their thoughts along new lines, and they were asked to elaborate on certain themes that seemed interesting and to express reflective and critical views. The study proceeded, from analysis and coding of parts of the data set, to developing a holistic understanding of the practices and views expressed by the interviewees. By applying thematic analysis, I classified the interview material based on a study of the previous literature on camera phones and mobile communication. I was also open to themes that emerged from the analysis or originated with the informants, but the analysis was predominantly theory based.

I will use pseudonyms when referring to the interviewees. They formed a quite knowledgeable group when it comes to photography and, in that sense, they were not representative of the general population. None of them were professional photographers. They were interested in taking and communicating with photographs, which, however, seems to be quite common nowadays, as is demonstrated by the popularity of posting photographs on Instagram and Facebook. A significant proportion of the

population in developed countries – and increasingly in developing countries as well – are potential photographers, as they have access to a camera in their mobile phone. In this sense, I suggest that individuals without specialist knowledge in photography can and do use their camera phones for mediating presence.

Kasper (aged 21) was specializing in digital multimedia in his college studies. He had previously worked as a photographer at a local newspaper and had also taken promotional pictures for music bands. Lotta (aged 21) was specializing in digital multimedia as well. She was a habitual photographer, describing her habits as 'taking pictures for fun, from parties and different occasions'. Like Kasper and Lotta, Joakim (aged 23) was specializing in digital multimedia. For him, photography was a hobby and he usually carried both a stand-alone camera and a camera phone with him. Mikael (aged 21) was a student of media culture. He had studied photography a bit but had not published photographs. Anja (aged 55) worked at a college as the director of a study programme. She took pictures of family and nature, as well as travel photographs. Kjell (aged 52) was a programme director. He photographed regularly, several days a week, and had also developed photographs by himself. Ulla (aged 35) was a teacher in the media technology department. She had the habit of building pictorial archives. She had archived plants and, at the time of the interview, was archiving photographs of VW Kleinbuses. Bengt (59) worked at a college as a research coordinator. He took a lot of pictures when travelling but also photographed his grandchildren when he met up with them.

The idea of mediating one's contemporaneous presence in a certain location by sending photographs has also been introduced in several other studies on camera phones. Even before the introduction of multimedia messaging (MMS) there were predictions that the 'sending of place' and the experience of 'being there' would become possible through camera phones (Kopomaa 2000, p. 96). In *Mobile Image*, a pioneering book on camera phones (Koskinen, Kurvinen & Lehtonen 2002, p. 78), it was concluded that digital images are used to share one's visually mediated experience, often in a very concrete way: 'Look, I am at a café with my friends! My, what a mountain of dishes I have to do! Now I am abroad, now I am parachuting!'

In 2005, Scifo (p. 373) wrote

the camera phone enables the multiplication of connections between different physical and social spaces, rather than the weakening of a 'sense of place' – even though mobile communication is often cited as contributing to the processes of disembedding experience from local contexts.

With its new visual potential, mobile technology emphasizes forms of experience that are strongly rooted in physical locations. The camera phone enables 'the doubling of place', or the translocation of place. In

photographic communication, the place or situation is transformed into the content of the communication itself (ibid., pp. 364–365).

In the study by Kindberg *et al.* (2005, p. 46), a subject described the photographic connection as ‘telepresence’, in that it made his absent girlfriend feel as if she were ‘here to see it’ (for similar views, see also Counts & Fellheimer 2004; Gai 2007, p. 205; Kondor 2007, pp. 27–28). One conclusion of Kindberg *et al.* (2004, p. 12) was that

Unlike text messaging, many of the images sent to absent friends and family were in fact visual evidence or proof of something having had occurred ... Proof of being somewhere or experiencing some event could be made more potent by showing when something was happening as well as what was happening.

Absence of the other motivates the communication of photographs (Rose 2003, p. 11).

In my study, the relation between photographs and place came up clearly in the discussion with the subjects. Joakim noted that by sending camera phone photographs one can ‘prove that you are somewhere’, in a ‘certain situation’. Photographs are not independent of place (Pink 2011, p. 9). According to Mikael, the reason for sending photographs might be that ‘It has been something funny you want to show some friend or you want to say *hi!* from a ski resort.’ He described sending many photographs portraying himself in specific places that acted as (indexical) proof of his presence there. For Mikael, the camera phone ‘brings maybe a new dimension to visual communication; you can send them [the photographs] wherever you are’. As for Bengt, he explained, ‘With this camera phone it is more to show where you are and what you do, and there is not so much aesthetic value in that picture.’ For him, a photograph can also act as an invitation or wish for the other to ‘join’ the place, as in ‘I wish you were here.’ According to Hjorth and Pink (2014, pp. 42, 45), ‘Camera phone practices provide new ways of mapping place beyond just the geographic: they partake in adding social, emotional and psychological dimensions to a sense of place.’ Locations become ‘somewheres’ through photographs.

The difference between text messaging, in mediating presence, and photographs was obvious. For example, during his trip to Japan, Joakim ended up in the middle of an enormously big crowd, and thus a photograph which says *hey check out the number of people and I’m standing right in the middle of them* would have been more effective than a text message sent from the same situation. Mikael observed that ‘Maybe the photograph helps to put the text in place and time, that now I am here and doing this, and this is the place where I am, whereas text messages are only, like, they could come from anywhere.’ This relates to how camera phone picture-taking and sharing can be viewed as part of a broader process of ‘emplacement’ (Hjorth 2015, p. 24).

Based on their descriptions, it is easy to notice how many of the photographs sent by the interviewees resemble postcards, especially ones sent during holidays. Kjell explained how he can send an 'instant postcard' with his camera phone: 'I want to say what I am doing now, this is my moment.' Bengt had bought a camera phone for his lady friend just before she went on a trip to Egypt, so that they could send photographs to each other. He had also sent photographs to his daughter from a trip to Italy and, in general, most of the photographs he had sent were from trips. Bengt explained that 'The longer the distance, the bigger reason there is to send [photographs]. You send postcards; these are instant postcards.'

Ulla described how she had sent a photograph to a friend from the hospital after breaking her leg. The photograph of the leg, all wrapped up, communicated, 'Greetings from hospital.' However, of the subjects in the study interviews, only Ulla explicitly stated that she did not feel like she was mediating her presence in a certain place through the photographs that she communicated from her camera phone. She rather liked to share what she was seeing. This can be regarded as an example of the use of mobile photographs as a form of television, or rather 'tele-vision'. This follows the meaning of *television* as 'vision at a distance'.

An aspect related to the communication of photographs as tele-vision is the 'synchronous gaze' (Villi 2015; Villi & Stocchetti 2011). Photographs do not necessarily represent face-to-face telecommunication; that is, the photograph does not inevitably include the face of the person sending the photograph. However, to achieve a sense of mediated presence, one does not have to see the other person, but rather feel that they are sharing the same view. This type of mediated presence is common to mobile communication in general. Ito (2004) makes an important point, in that the metaphor in text messaging is side by side, in contrast to the face-to-face modality of a telephone conversation. What mobile photographs add to the side-by-side presence of text messaging is the aspect of visual perception. A photograph offers both an interpersonal, shared experience and a common view to the same world through the photograph.

The idea of the synchronous gaze surfaced in several of the study interviews. The use of the verb 'to show' was exemplary of this in the descriptions of the camera phone communication practices. Joakim remarked that '[With a photograph] you can show unusual or strange things happening at the moment they happen, and you can send a picture of that to a friend or somebody: *hey, something like this happened right now.*' Bengt commented

I think that those [photographs] I would send from the phone have to do with what's going on, the feeling which is there; there is a connection that *I'm here now watching this ...* I can take a picture and show that *hey, now this is going on ... share the experience.*

Lotta noted

A text message is a simple way just to say one thing you have to say, and if you put a photo in it, it's probably because it is something fun or good-looking, like a view or something; you want someone to see what you are seeing [in the place].

Conclusion

In this chapter, my intention has been to join the study of camera phone photography with the discussion on location-aware mobile communication. Based on studies focusing on the communication of camera phone photographs, including my own qualitative study, it can be claimed that there is often a connection between the physical setting and the act of photographic communication. The 'phonespace' is appended with visual peeks into the changing locations of the communicators. The 'photospace' – the telecommunicative space of camera phone communication – is thus more location-bound than the phonespace.

The main argument of the chapter is that photographs can emphasize the relationship between mobile communication and location by incorporating the location to the communicative act. The sender of the photograph can provide a visual notion of her or his existence in a certain location, the *I am here now*. The indexicality of the photograph serves as the guarantee of presence. Thus, the location is not just a point of communication and mobile communication is not only about engagement in the phonespace. Accordingly, photographs communicated from a mobile phone can provide, in addition to social presence, the sense of being together, also physical presence, the sense of being there. The photograph re-localizes the act of communication. This is in contrast to verbal mobile phone communication, where the actual location of communication often has very limited significance and is not necessarily known to the other communicator. The location is often mainly manifested only as background noise.

When using the camera phone to capture and communicate a photograph, one's presence in the place can be the purpose and motivation for the act of communication, similarly to when a user checks in on Swarm. Communicating a photograph and mediating presence by checking in are both examples of location-aware networked interaction. Interestingly, the GPS features in mobile phones, that enable many location-based services' applications, can further amplify the physical location in photographs, as when photographs are supplemented with location-related metadata. In Augmented Reality (AR) applications, the camera apparatus in the phone is given an additional role when, in addition to capturing images of places, it mediates location-based information to the user. When utilizing AR applications, the camera is a window – not a window to another place, as in tele-vision, but a window loaded with local information. In this sense,

the screen in the camera phone is not only a viewfinder but also an 'information finder'.

Location-aware mobile technology forms a context within which to explore new and old models of communication, connection and communality. Mobile and networked interactions do not occur only in the phonespace, but they are also increasingly connected to the physical location. The camera phone is one of the first location-based applications in mobile communication and certain practices of photographic communication act as a forerunner for the location-aware technologies, services and mobile apps that have been introduced in recent years. In my view, the concept of mediated presence particularly serves as a valuable framework for studying the practices that are emerging in the field of locative media. Naturally, mediated presence does not relate to all forms of location-based mobile use, but clearly it relates to the use of Swarm, for example, and other apps that emphasize the sharing of place. A check-in can act as authentication of one's 'present presence' and, at the same time, links the users to their friends or followers. Very much in the same way, a photograph, communicated from the place photographed, can authenticate the contemporaneous presence of the communicator, share the place and establish a connection.

Note

- 1 In 2014, Foursquare introduced the spin-off app Swarm to include the check-in feature previously offered by the Foursquare app.

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11 Photographic places and digital wayfaring

Conceptualizing relationships between cameras, connectivities and transformed localities

Sarah Pink

Introduction

In this short chapter I elaborate on two concepts which I have used in my existing work, and that I would consider to be part of a theoretical 'tool box' that digital media scholars might draw on: place and wayfaring. These concepts, although they were not originally developed in the context of media or photography studies, are useful for thinking about how digital photography is at once technological, virtual and material, often played out through human activity in the world, and also part of the actual environment in which we live and act. However, we can also take this further to join up the concepts as a way to consider a framework through which to think about the contemporary digital, material and representational environments in which we play out our lives. This framework also brings together the different areas explored in the chapters in this part of the book, which deal with different themes – transnational families, activism, location and the online/offline – but collectively share some interests across the themes of how the everyday activities through which digital photography becomes part of our lives, the relationships in which these are implicated and the environments in which this happens, and how these environments and relationships are constituted and experienced.

Place

The concept of place is well established across a range of literature, and has been used most consistently in human geography, as well as in anthropology. For me, the most interesting works in this area are by the human geographer Doreen Massey (2005) and the anthropologist Tim Ingold (2008), because they respectively call our attention to the politics and phenomenology of place (Pink 2012). I do not repeat their arguments in detail here; however drawing on these ideas, I would reiterate two points that are important to account for when discussing digital photography.

First, that place should not be confused with locality; locality can be part of place, and of course usually is. In the case of digital photography and its online sharing, for instance, photographs might be produced in localities, they might represent aspects of localities and they might also be viewed in localities. However, if we are considering the digital material world in which digital photography is made, shared and given meaning, then we need to recognize that the configurations of place we are concerned with might be constituted by several different physical localities, as well as different digital platforms, technologies and social relationships.

Second, I emphasize that place is never fixed, but continually changing. Therefore, the configurations of place in or through which we consider digital photography need to be understood as contingent and continually slipping over into an, as yet, unknown future. Such contingency shapes not only the ways in which we find and share images but also the ways in which they are made. As I have also argued, drawing on this work elsewhere:

Images, thus, are not 'of' places or things. They cannot be, as they are not independent of place and they are contingent on encounters with things. But rather, they are inevitably and unavoidably in places: they are produced by moving through and not over or on environments, and they are not stopping points so much as outcomes of and in movement. This means making a shift in thinking from the commonsense idea that a photograph represents a static moment and thus is a photograph of a place, to the idea that it is in place and in movement.

(Pink 2011a, p. 9)

Recently, some media scholars have started to engage with place as a way of thinking about the way that media become part of the global and local configurations of things and processes in which we live (Couldry & McCarthy 2004; Pink 2012) as well as a way of thinking about the situated nature of our everyday uses of media (Moore 2012; Pink & Leder Mackley 2013) and in relation to mobile media (Wilken & Goggin 2012; Pink & Hjorth 2012, Hjorth & Pink 2014). The important step that these writers take, beyond the initial theoretical discussions of place that have been played out in academic disciplines like human geography and anthropology, has been to begin to explore how the digital and material elements of our worlds are equally part of the same environment. Place, as I have noted above, is not simply locality, we do not need to attend to digital media to acknowledge this, yet, when we do, we are able to use a concept of place through which to consider how the digital and material (which refers to the online and offline, but more than that) come together. However, we should also keep in mind that we cannot understand the relationship between the digital and material without also accounting for other elements of what Ingold would call the 'weather world' (2010), including

invisible and intangible elements, to form the environments of which we are also a part. As Tim Ingold reminds us

We are these days increasingly bombarded with information about what is known as 'the environment' ... that we are, I think, inclined to forget that the environment is, in the first place, a world we live in, and not a world we look at. We *inhabit* our environment: we are part of it; and through this practice of habitation it becomes part of us too.

(Ibid., p. 95)

Ingold was not writing about digital photography but, as is often the case, his comments are insightful for generating understandings beyond the contexts he writes about. Here, I suggest taking the idea that we are part of and inhabit our environment as an invitation to consider how we do so with and in relation to digital photography and technologies for producing and sharing them. Here, I do not simply refer to digital content but rather the practice, sharing and viewing of digital images and how this is related to the way in which digital is part of our everyday lives and the invisible architecture of our homes and the streets we walk along.

Digital wayfaring and moving images

The above point brings me to the second concept I wish to discuss – the notion of the digital wayfarer. This concept is discussed in more depth in my writing with Larissa Hjorth (Hjorth & Pink 2014), and used more specifically to discuss camera phone photography. The concept, however, has a wider application as a way to understand how people can be thought to weave their ways through the digital/virtual and material/physical worlds that I have described above. A theory of place focuses on the ways in which configurations of things and processes come together, and the relationality between them, without necessarily privileging the human subjects who are part of place. The concept of the wayfarer, as I develop it here, instead enables us to focus precisely on the question of what people are doing, as they move through the world – that is to put the human subject at the centre of the analysis.

The concept of wayfaring was developed by Ingold in his work on the line (Ingold 2007), to discuss the ways in which we might think of people meandering through the world, attending to what they find on their way and engaging with their environments in doing so. The notion of the digital wayfarer (Hjorth & Pink 2014) developed this to apply the same idea to the ways in which, when we use camera phone photography, we are weaving our way through a digital-material environment. Camera phone photography, as we discuss, is particularly interesting in this respect because people might take photographs of their environments and/or themselves in them, upload them and engage with people in relation to them

online. We see how the digital and material elements of people's everyday journeys and activities are entangled in the processes that camera phone photography involves (*ibid.*). As the chapters in this part show, digital images are made as people are active in and move through the world. As I have argued elsewhere, photographs can be understood as being made as we move forward through the world (Pink 2011b).

It is not only people who move but also images (Pink 2011a). We see examples of this in the chapters in this part of this book, for instance in the case of images related to a contested political event as much as in the case of the photographs of transnational families, or photographs that connect people over geographical distance through web platforms. As images move through the world, they become interwoven in new stories, and those people who engage with them, take them up and journey with them some way, and they become part of new 'stories' (to borrow a term from Massey 2005) as images are implicated in and/or move through different configurations of place, sometimes being pulled in by human actors.

The images in this part of this book are an excellent example: they have not only become part of the stories that the authors of the chapters have narrated to us in their descriptions, but now they also sit with us as readers in a place where they have become academic photographs, entangled not only in the stories that they have emerged from, but also with those of the academic argument made in the narrative of each chapter, and also in the stories that we as readers tell ourselves with them. This scenario is akin to what I have called the 'ethnographic place' (Pink 2009). While I have developed this concept specifically as a way to discuss how sensory ethnographers bring together our research materials, arguments, stories, experiences, writings and more, it can equally be used to think about academic writing about qualitative research more generally. The concept of place can, likewise, be used to consider the configurations of things and processes that come together in academic work, which is also something that is open to change. Our writing and photographs never stop at the texts we present them in, but rather they will always go on new journeys with the readers who pick them up and intertwine their reading with our words and images as they move forward through the world together.

Moving places

Theories of place and theories of movement are commonly brought together. The work of both Massey and Ingold, discussed above, suggests that place is constituted in movement, by moving people, things and processes. Above, I have suggested that we understand digital photography as part of places. I have argued that digital photographs are made and experienced by people in movement. I have also proposed that we see digital images as themselves in movement – as digital artefacts that are not static but always going forward through the world and becoming part of the

stories of the people to whom they become related. Digital photography is one of the everyday life, professional and technological practices and activities that are entangled with everyday life, and both photographic technologies and practices become increasingly ubiquitous. It is part of the way in which our worlds are made, experienced and given meaning. The task for us, as digital photography scholars, is to unpick the ways that this is happening, and what it means.

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Part III

Camera as the extension of the photographer

Camera as the extension of the photographer

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12 Exploring everyday photographic routines through the habit of noticing

Eve Forrest

Prologue

The first time it happened to me was whilst I was out with a group of photographers on a quiet Sunday morning; the city streets of Newcastle-upon-Tyne were deserted. I was feeling uninspired even though I had two cameras with me: a newly purchased toy Holga and my trusted 40-year-old Pentax which I wore around my neck. Looking around me, I could see little that I thought was worth photographing. However, one of the photographers beside me suddenly stopped still, then, crouched down. I observed them at work as they moved around: making micro adjustments to their feet, fingers, shoulders and head movements, getting their body in the best position to take the picture. They lingered over what they had seen for a full five minutes taking a variety of different pictures. This episode proved not to be unusual that day: in the few hours I spent with the group, the same thing happened numerous times with other photographers. Despite standing in close proximity to them, what had grabbed their attention had passed me by completely unnoticed. My camera was little used that day and my notebook had one loose scribble underlined: *why can't I see the same way as them and when do photographers start seeing things?*

Introduction – what catches the thread?

Often, when discussions turn to describing the creative process, it is said that something can *catch* the imagination, much like when the singular thread from a scarf snags itself on a nail and begins to unwind itself from its tight weave. In much the same way, imaginative thoughts are pulled through and out of the front of our consciousness, housed deep within the varied interlacing routines of everyday thoughts. As Abraham, McKenna and Sheldrake (2001 p. 36) attest 'motivations in the ordinary psychological sense are not pushing from behind but pulling from ahead ... somehow the system ... is subject in the present to the influence of a potential future state that hasn't yet come into being'. Quite unexpectedly, something or someone can catch the attention. When this happens, the notional thread

can only be grasped and, if desired, the holder follows the potential trail wherever it can lead.

Many have asked, in the course of my fieldwork, what separates a photographer from a person who just carries a camera? What is the nature of that label? It is a difficult weave to unpick. Having spoken with numerous photographers over the course of many years, the common theme in each conversation was the transformative powers of carrying and using a camera every day, be that film, digital or on a mobile phone. They remarked on how they saw the world differently, markedly so from the others around them that did not habitually take photographs. They also noted that after prolonged use, even when they did not take a camera with them, they still saw potential photographs everywhere within their environment. Thanks to the presence of the camera, the imagination of the photographer is often unravelling, on any possible number of everyday hooks.

This chapter gives consideration to how and why this particular kind of imaginative process, or what I term here *noticing strategies*, happens. The phenomenon of noticing is first discussed by David Seamon (1979) in his seminal work *Geography of the Lifeworld*. Seamon (2000, p. 161) later on writes that 'it is impossible to ask whether person makes world or world makes person because both exist always together and can only be correctly interpreted in terms of the holistic relationship, being-in-world.' Using *Lifeworld* as the primary touchstone I will consider the core experiential qualities of photography practice and discover, phenomenologically speaking, how the photographer and their being-in-the-world is affected by the presence of the camera. This work draws from other disciplines too, that whilst initially appearing to be unconnected, are tied together by what could be loosely bracketed as phenomenological threads. The anthropologist Tim Ingold and his ideas on perception (2000) and movement (2008) within everyday life are used to consider noticing strategies online. The work of Terence McKenna, 'anarchist metaphysician' (St John 2011, p. 207) is also worthy of further exploration here, more specifically his work relating to creative consciousness and ontology within everyday life.

The hitherto mentioned conversations were part of a longitudinal ethnographic study of photographers based in North-East England, more specifically, the cities of Newcastle-upon-Tyne and Sunderland. The photographers came from a variety of backgrounds and ages: the only links were geographical location and Flickr, from which they were all enlisted. The photographers presented a mix of abilities (although the traditional labels of amateur and professional, I would argue, are largely outmoded on Flickr and other photography platforms: see Forrest 2013). Often it is now difficult to tell between the quality and styles of images that used to make differentiation more straightforward. I interviewed more than 20 photographers in various locations, both online and face-to-face.¹ At least half of these photographers were then accompanied on an hour-long 'walk and talk' around their local cities of Newcastle-upon-Tyne and Sunderland.

This approach, also known as the guided tour technique, has its roots in what could be widely described as a phenomenological attitude towards thinking about everyday practices such as photography, giving opportunity for the researcher to 'observe their informants' spatial practices in situ while accessing their experiences and interpretations at the same time' (Kusenbach 2003, p. 463). This method emphasizes the 'call to lived experience' (Landau 2004, p. 111) of the photographer, rather than the images they produce, meaning a fuller exploration of the multifarious *doings* of everyday photography, are made possible.

Geography of the Lifeworld

Whilst *Geography of the Lifeworld* (referred to hereafter as *GOL*) has been utilized by a few authors in diverse areas such as urban planning (Wunderlich 2008) and media studies (Moores 2006, 2012; Moores & Metykova 2010) the work is still relatively unknown and underused. It is therefore worthwhile briefly outlining the pertinent themes of *GOL* and how they relate to photography practice. *GOL* is a study of the everyday complexities 'and inescapable immersion in the geographical world' (Seamon 1979, p. 15) via the 'body ballet', a set of integrated gestures, behaviours and actions that sustain a particular task or aim (Seamon 2006). These different forms of gestures and actions connect with the practice approach, as outlined by Lehmuskallio and Gómez Cruz in the introduction to this volume. They argue that the visual is more than representation alone. Practice theory facilitates an exploration into the various routine performances and communicative actions with a camera: a particular dance set to the rhythm of everyday life.

Seamon particularly underlines the impact of habit, drawing attention to factors that structure these various interactions. The distinctive approach of the original text was based in a branch of human geography that came to prominence in the mid-to-late 1970s, through authors such as Relph (1976) and Tuan (1977), which 'shifted analytical focus from social space to lived-in place, seeking to supplant the "people-less" geographies of positivist spatial science with an approach that fed off alternative philosophies – notably existentialism and phenomenology' (Hubbard 2005, p. 42). Seamon particularly utilized the phenomenology of Merleau-Ponty to investigate these everyday actions up close, in relation to the body-subject. Using the technique of themed 'environmental experience groups' he asked his participants to think and observe their patterns of movement and habits in their everyday environments via a theme assigned to them each week. Moores (2006) summarizes that 'Seamon's groups can therefore be thought of, in a certain sense, as having engaged in consciousness-raising ... through the expression of at least some of these tacitly known things that enable the skilful accomplishment of everyday practices'. It is through the outcomes of these group meetings that Seamon persuasively

argues that our relationships with place and habitual routines are highly complex and not merely a symptom of automatic reinforcement or a condition of set thought processes. Instead, Seamon (1979, p. 40) believes that everyday interaction and movement 'arises from the body' which is 'at the root of habitual movement' (p. 41). Seamon (p. 55) describes the nature of the everyday body ballet, stating 'Basic bodily movements fuse together ... through training and practice. Simple hand, leg and trunk movements become attuned to a particular line of work or action and direct themselves spontaneously to meet the need at hand.'

The metaphor of dance is particularly useful for thinking about how photographers can vary their everyday movements. When out in groups they waltz in-between one another, so as not to disrupt the image taking of others. Often, of course, they dance solo and, depending on the camera used, they do not just use their feet. They bring their eye close to the viewfinder or hold the screen at arm's length, stretching their muscles out or pulling their elbows in tightly to their body. Even when they are 'still' and watching the world with the camera in hand, they continue to make micro adjustments with their weight, shifting from one foot to the other. Sometimes they even go *en-pointe*, tip-toeing around their subject to gain a better perspective.

These actions, which can be viewed as part of a complex interlacing between body and camera, are being further advanced by new technologies and hardware. Where cameras used to be extensions of the hand and eye, they can now also be integrated into the body through various wearable means, such as clips, mounts and harnesses on the chest (for example GoPro cameras and accessories and the Apple iWatch). Photography practices have, then, become both embodied and an embedded part of everyday life (see Favero, this volume).

One strand of this intricate, embodied ballet is highlighted in Chapter 14, which Seamon (1979, p. 108) names 'Noticing and Heightened contact'. He defines noticing in the following way, 'A thing from which we were insulated a moment before flashes to our attention. Noticing is self-grounded or world grounded ... Incongruity, surprise, contrast and attractiveness (or its opposite unattractiveness) are all characteristics that activate world-grounded noticing.' Whilst Seamon does not specifically tie noticing to image-making practices, I wish to demonstrate, throughout this chapter, that his definition, above, of the particular experiential qualities of noticing, aligns closely with that of the photographer. Importantly, the only instance when photography is mentioned anywhere in GOL is in the context of noticing. Seamon (*ibid.*, p. 109) writes that

One group member, pleased with the photographs he had done, spent an entire afternoon taking more pictures. 'I was noticing more than I usually do', he said 'and it had something to do with the fact that the photographs had come out so well.'

Before moving on to wider discussions on everyday photography, it is interesting, here, to consider the other area that Seamon associates with noticing: heightened contact, particularly in the context of Terence McKenna's statement that more attention should be paid to the 'felt presence of direct experience' within everyday life (McKenna 2012). The unmistakable transcendental tone of the language Seamon uses in this chapter has more in common with that of McKenna (1991, 1999) and his psychedelic, spiritual and ecological explorations. Seamon (1979, pp. 111–112) notes that through heightened contact 'a feeling of peace and harmony with the world' is possible, that participants noted that everything 'feels more real' in that state of mind and that the person 'feels joined and akin to the world'. This also echoes the writing of Graham St John (2004, p. 21) who, in his exploration of raving, quotes Spurgeon noting that the space of the dance floor becomes 'the space of "awakening ... [raving] is an experience that amounts to rapture"'. In a similar way, Seamon argues that heightened contact involves 'the physical environment beyond the person' (1979, p. 113).

All these aspects are pertinent to discussions relating to everyday photography, particularly the unfolding of movement and space within the urban environment and how photographers engage and reflect on their everyday surroundings. It is worth noting here, too, that the type of equipment used (for example a phone or DSLR camera) as well as the publishing platform that photographers decide to put their photographs on to (such as Flickr or Facebook) may also impact on the different noticing strategies of photographers and, perhaps, with different results, that are discussed within this chapter. All the photographers here used primarily DSLR or film cameras when out 'doing photography', but by the very nature of their primary function they also carried their mobile phones with them too. When they are outside, walking around with the camera, the photographer is acutely aware of the surrounding environment and awakened to its potential. They are aware of themselves, the presence of the camera in their lives and of those who will be looking at the images once produced. Their interaction with both machine and the world around them can often lead to what Abraham Laslow would call a 'peak experience' of creative energy. It is to these various imaginative incarnations that this chapter will now turn.

The various habits of routine noticing

'Who would watch empty streets? Such situations are neutral backdrops and normally do not capture the experience's attention' (Seamon 1979, p. 106).

When I am going out in Sunderland, I go where I think I am going to get good photos ... you see the world totally differently ... you are keeping an eye out.

(Participant M)

To respond to Seamon's quote, it is the photographer who is always watching whether the streets are empty or filled with people, if the buildings are in light or shade, whether it is warm or damp, afternoon or night. As the hundreds of pedestrians walk along the same paths, allowing the neutral backdrop of the cityscape to merge with their everyday activities, the photographer is noticing. The focused inattentiveness of the urbanites is one part of their natural attitude, like the noise, dirt and architecture which inform their everyday life (Bull 2007). However, photographers here see the city as a space offering an infinite number of sensual opportunities. They interact with its space in complex and enhanced ways, actively engaged in looking for noticeable scenes, people and places. Photographer A2 noted

I am always thinking with photos in mind; it is kind of an obsession. Once you start taking photographs at a regular level you start to see things through a camera, even if you don't have one.

The camera, then, does not always have to be in hand to instigate noticing. Even when they are without one (although this is rare), photographers are frequently thinking and acting as if they have it with them. Instead of the photographers' awareness 'advancing and retreating like the actions of waves on shore' (Seamon 1979, p. 103) the effect of the camera on their everyday interactions and movement is more permanent. Another photographer (U) also explained

I walk around, I carry the camera everywhere – that is the best thing about digital, you can take the tools of your trade with you – and, yes, spotting things becomes an annoying habit when you are with other people.

The annoying part inferred to here is that, when photographers notice things, they frequently need to perform a number of different actions. They, first of all, stop, get the body into position and work out the picture they want, pressing the shutter release, whilst often trying different angles. They then review the pictures and then constantly repeat this process until they are happy, by which time the companion they were out with has walked farther on, or complained that they are bored. Other photographers told me similar stories:

My kids and my wife think I am mad. I will be having a cup of coffee and start staring at something and move my head a bit because I will be interested by a shape, the way the light has hit something, or a reflection, whatever.

(D)

In the above example, noticing happened in an unanticipated way, as, although the photographer was not expecting to find a picture in a coffee

shop, something subtly changed in the light or movement. Reacting to changes with small micro-movements, the body instinctively adjusts, as movement 'unfolds instantaneously in one smooth flow' (Seamon 1979, p. 109). Merleau-Ponty (cited in Carman, 2008, pp. 110–111) notes that 'intentionality of perception depends crucially on the normativity of body schema ... we have a *feel* for the kinds of balance and posture that afford us a correct and proper view of the world'. In the case of the photographers here, this view is subtly different from the others that they are with. Another photographer explained that

I'd say I am always photographically aware of my surroundings, carrying a camera or not. It is rare that I am not carrying a camera of some sort; if I was without a camera I would wish I was carrying [one].

(P1)

What does it mean to be 'photographically aware' of surroundings? It differs from photographer to photographer, depending on their preferred subject, but what is important for each of them is habitually and intentionally paying attention to the world around them. The end goal is to find an interesting picture within the overlooked details that are scattered around the everyday environment. Seamon (1979, p. 109) notes that 'training and interest providing some control of noticing ... but different people notice different aspects of the same environment'. Certainly, this was true of many of the photographers, as one explained:

If you are carrying a camera around with you, you are always looking for something to photograph; it's a strange situation. You don't consciously look for a photograph but, when you are wandering along, something will take your eye and you will look at it, move around, you look at it from different angles and all of a sudden you will see a photograph and you will say *I like that, I want to take a picture of it*.

(P2)

Interestingly, the description given above implies that the noticing cannot be forced, even when they carry the camera. The common frustration of the photographers I spoke to is that wandering can often be fruitless, they do not notice anything of interest and nothing comes forth. In this regard I would like to say more about emotion, noticing and photography. When questioned about mood or emotion when taking the camera out with them, participants seemed hesitant to attribute directly one factor to their style of photography on a specific day. However, emotion did have an impact on their photography in general and their noticing strategies in particular. M told me that 'if I am in a bad mood, I won't go out and if I am in a good mood, I will take more risks and go out'. This sentiment was also echoed by other photographers, too, who commented how negative emotion made an impact on their photograph taking:

Yes, I find that if I am not feeling well or in a good mood then I don't take photos. I have to be positive about stuff. I am an upbeat person anyway but if I have a lot on my mind then I don't tend to take as many photographs.

(C)

I have to be in the mood for it. The summer is a bad time for me because of the light [he prefers night photography]. I might go somewhere and something might happen and it will put us off and I will be in a bad mood and I know I will not take decent photos, so I won't bother and just write it off, but definitely have to be in the right frame of mind.

(A2)

Negative emotion does affect the bodily rhythms and processes of the photographer, though it is more difficult to articulate exactly why. One possible reason is that, once fully absorbed in the action of photographing, the mind and body are knotted. Soon, full concentration is given over to photography and the individual becomes entwined in the world, their habitual movements flow, in turn creating more noticing opportunities, leading to further absorption, and so on. In regard to habit, Seamon (1979, p. 40) comments that 'movements occur without or before any conscious intervention', so if there is a negative feeling of frustration, this self-awareness somehow blocks this process from working. Instead of flowing, the body's rhythm stutters as concentration is lost and noticing becomes more difficult. Another responded in different ways when asked about emotion and their photography:

I feel bad if I don't go and take photographs and it can affect my mood if I go out and I might spend all day taking photos and it is the last 10 minutes and I get the unexpected shot. I think people think that it's dead easy but it isn't.

(A1)

The notion that photography is easy to do is a common one within popular discourse.² Noticing could also, then, be viewed as a way of the photographer consciously separating themselves as different from (in their opinion) other, less dedicated photographers. One told me

I try and notice things other people wouldn't,

(D1)

which is a common trait amongst many photographers. The 'other people' talked about here were not just those who took the occasional photograph with their mobile phone but also other keen amateurs and professionals.

The skill photographers most appreciate in the work of others is not just the ability to capture a picturesque landscape or portrait, but to see *something* where they have not seen *anything* and create a photographic opportunity through noticing alone. During a walk around, one photographer commented

the thing on Flickr meets [where online groups meet up offline] is that there is an unofficial competition to see who can get a shot that no one else gets. It is a challenge ~~to get something~~ different to what everyone else is doing.

(D2)

Building on earlier discussions relating to the body schema, it is worthwhile giving brief mention to Bourdieu's ideas surrounding habitus,³ a blend of 'conditionedness and spontaneity' (Carman 2008, p. 219) in relation to noticing. Tim Ingold (2000, p. 162) explains habitus as 'how people acquire the specific dispositions and sensibilities that lead them to orient themselves in relation to their environment and to attend to its features in the particular ways that they do'. As part of their skill set, photographers become accustomed to looking, in order to find interesting things to photograph, which has subsequently formed into a more solid habit. The photographers below all identified this link:

I do go about and think, what would be a good image here? So I will look at a street like this ... look at the patterns of buildings, the repetition. What I will really look for is a person on the street because that is what I find interesting ... you do look for an image.

(A1)

Before I got a camera, I never looked up so much but there is so much hidden detail that people aren't really aware of around [Newcastle] so I quite like that. It wasn't until I got the camera that I started looking all over at anything, all sorts of different things.

(D3)

My eyes are everywhere, looking about. I am an inquisitive sort if I am walking around a town. If you look above the shops and on the pavements there is stuff there. I went through a spell of taking pictures of the corners of things ... especially when you are cross processing, that enhances the colours of things. I don't plan it, though, I just carry [the camera] and see what happens.

(C)

For Bourdieu (1977, p. 87), habitus lives in the type of noticings described by the participants, above. It is implicit within spatial and body awareness

or, in Bourdieu's terms, it can be 'a way of walking, a tilt of the head, facial expressions, ways of sitting and of using implements – all of these, and more, comprise what it takes to be an accomplished practitioner'. Re-enacting the traces of previous bodily movements, the photographer does not necessarily know what they are looking for until they find it; their world unfolds in front of them and they react to it as part of their body's wider intelligence.

What am I *drawn* to? Noticing strategies and movement online

Although the main discussion, here, revolves around photographers and their strategies with the camera offline, noticing also has an important role to play online, too. The folksonomy of Flickr leads to a random ordering system of images, the exploration of which is more akin to a virtual form of what Ingold (2000) terms 'way-finding' (see Pink, in this volume). Users find their own paths to the people, groups or pages they frequently visit on the site. For Ingold (2000, p. 220), places 'exist not in space but as nodes in a matrix of movement ... ordinary wayfinding more closely resembles story telling than map-using'. Ingold has never written about movement in and around online environments. However, his concept of wayfinding very much lends itself to platforms such as Flickr and how users get around or find their way in the space. Discussing navigation and dwelling in physical space, Ingold (*ibid.*, p. 230) believes that people's 'knowledge of the environment undergoes continuous formation in the very course of their moving about in it' so that they 'know as they go' (p. 228). In more ephemeral places, like Flickr (Van Dijck 2011), the knowledge of where things are located, coupled with the unexpected images that can be found in the process of wandering about, means that different paths are continuously forming and disappearing. To this end, Flickr is not a network of points, but a *meshwork* of interactions, a notion also utilized by Pink (2011, this volume), too, in relation to other related photography practices.

Noticing also has a role to play as this part of this navigation strategy. When I observed the photographers⁴ accessing their Flickr photo-stream and moving around the site, many used a similar terminology in previous descriptions of how they used the camera and wandered around the city. As they wandered through, clicking from image to image they told me, frequently, how they were 'drawn to' particular photographers and their subjects. They also quickly became absorbed in looking at the images, only making micro-movements with their eyes, and their index finger slightly twitching on the touch-pad to move the cursor. When they saw an image they were interested in, they would often go on to explore that photographer's page, looking at the rest of their images. In other instances when photographers were on Flickr, another repeated movement was that of

continually going back and forth between different images, their personal page and groups. Sometimes they would open up multiple tabs to aid these movements and then begin to comment on, favourite or add the chosen photographer as a contact by utilizing different tabs. One participant (T) used continuous quick movements of scrolling down the page, hitting the 'favourite' button multiple times so he could find his way back to the work that 'caught his eye' in the first place.

This happened with other users as well. During our session, one photographer (A1) refreshed the main Flickr 'Explore' page around 20 times, quickly scanning the page and clicking the 'reload' button. In this instance, the hand, eye and computer were synchronized in continued repetitive movement and, with both participants, there was a definite form of concentrated distraction in their looking practices when exploring Flickr. Other photographers also became absorbed in looking at images. This distraction forms part of the complex interaction with the images, and they scanned the screen for an image that they were pulled into, in a similar fashion to the earlier *noticing* strategies. Like wandering in the city, often, when the participants began to explore Flickr, there was no set destination or final place they wanted to get to, so the threads began to weave around one another as they moved across the pages, creating a complex virtual tapestry. This movement was punctuated by noticing, or being drawn to, particular images.

The pages that they navigated around were, of course, deeply familiar to them and they revisited the same places on Flickr in a similar way to their favourite urban sites. Although wandering online, to an extent, is unplanned, ultimately these users are repeatedly exploring the same places (groups and certain features), revisiting the same sites on every visit. There were many overlaps in photographers' online and offline wayfinding strategies and the way that photographers know, as they go around Flickr, has parallels to their offline walking. The exploration was built from routine encounter, similarly, in their physical explorations of city spaces. Participants did not tend to seek out the unknown parts of the city, revisiting the same territory, always alert to new opportunities brought about by having the camera in their hand.

Horan (2000, p. 23) writes that 'digital places are new leverage points for creating new experiences and relationships that will profoundly redefine our experiences of physical space'. Thanks to the different and varied interactions on Flickr, photographers are also attuned to noticing different things and people around the city, too, inviting other photographers to join them in the process. Flickr has, additionally, become part of the everyday routine for many people, and their habits with the camera are also partially informed by the images they view on the site.

Watching and waiting

Thus far, I have highlighted the complexity of the photographer and their habit of noticing offline and online. However, there is another facet to this strategy that is important in their engagement with the world: watching. Ingold (2000, p. 25) notes that 'perception is grounded in an act of attention' so it is through watching that photographers begin to notice their surroundings or, to paraphrase Ingold (p. 265) it is watching which 'transforms passive looking into active noticing'. The participants in this study stated that they not only enjoyed watching, it was a part of their general strategy when outside with the camera. Some identified that watching was part of their integral personality:

I am a people watcher I was always interested in what was going on before [without the camera] though!

(R)

It's getting into [the subject's] head – I have always been a people watcher, even when I go out and have a drink. It is a natural curiosity about the world in general.

(G)

Another expanded on this idea and told me that photography can be

Quite voyeuristic ... I like people watching. I could do that all day, watching people pass by. Sometimes you can tell a story with a photograph, you can choose to show the environment or knock it out of focus and choose to show somebody ... I just find it very, very interesting to look at a photo and think *who is that person?*

(A1)

Seamon distinguishes watching from other forms of looking and describes it as 'a situation in which the person looks out attentively upon some aspect of the world for an extended period of time' (1979, p. 105). The photographers, here, detailed how they were pathological in their watching, always on the lookout for their next photograph. When I was out with one participant (R1) he commented that he 'hangs around and waits' so he can see where you get a good view of where people are. Later, when discussing his Flickr photo-stream images, he also noted that he often watched (from a distance) to get the photographs he wants. Seamon (ibid., p. 106) believes 'watching establishes an extended span of attention between person and place. To watch is to pay attention at length to the world at hand – to have one's interest occupied as mutually the world receives that interest.'

Replace the verb 'watching' here with 'photographing' and soon ideas can begin to be formed on the kind of interactions and exchanges which take

place between photographer, camera and city. The process of watching also becomes involved in the spontaneous planning of a picture, an evolving process between body, imagination and vision. The photographer makes sure that they move their body correctly, anticipating an upcoming shot.

When you do have [your camera] you just kind of take things a lot slower, walk places sometimes or even try and look ahead and stand, or I'll move over there, so then that might be something interesting. You can let everything go over you ... sometimes you have to almost create a shot, make sure you get into a position if you can see something developing. Sometimes ... I will walk past someone faster than I normally would and then come back [taking their picture] so they don't notice.

(L)

In this example, the combination of routine watching and movement leads to noticing opportunities. It also points to the idea that noticing is not just a random act but something over which the photographer tries to exert some control and, as Rupert Sheldrake reminds us, development of any kind of skill does 'involve interplay between habit and creativity' (Abraham *et al.* 2001, p. 3). Through watching, noticing can then at times be pre-empted, especially when the photographer repeatedly revisits a place they know, so, to a certain degree, they can expect what to find. Watching is important to photographers because it means they are always tuned into the possibility of photographic opportunities, taking advantage when a chance presents itself. When discussing a picture that he liked on his photo-stream, T told me

I just like all the brick and stone, so many lines. Once I sharpened it, all the lines popped ... there is really nice stuff everywhere you look, all the time, to me at least. It is just a question of translating it into the camera.

For the 'accomplished practitioner' (Ingold 2000, p. 162), the physical presence of the camera assists noticing, too, the weight of the machine on the body leading them through a familiar series of habitual movements as 'your perception of objects is already structured by your body and its sense of its own possibilities' (Carman 2008, p. 106).

Conclusion: noticing as *presence*

'The camera is an instrument that teaches people how to see without a camera' (Dorothea Lange in Meltzer 1978).

I asked, at the beginning of this chapter, what separates a photographer from a person who carries a camera. What is the nature of that label? The category of 'photographer', thanks to numerous technological changes, is

now particularly slippery, in an age when most carry some form of image-taking device in their pocket (Hand 2012). However, one could ask the same of other, simpler technologies. If one carries a pen, does that automatically make one a writer? I wear training shoes everyday, but does that make me a runner? The answer to these questions is both 'yes' and 'no'. If I want to run in my trainers, they are designed to support my feet, just as the pen helps elucidate my thoughts on paper. However to be a 'runner' and a 'writer' is a classification beyond the technology that facilitates that process. In this regard, I believe that it is the same for photography. To be a photographer is not always necessarily to do with the *quality* of the images but, instead, the way that the photographers perform photography with their body, their commitment to movement, the habit of noticing and how, through this habit, they observe the world. Tomi Hahn (2007, p. 163), in her work on Japanese dance, considers the knowledge that resides in the intelligent body: that which gives the dancers presence when they perform for an audience. She writes 'I am not certain it is possible to definitively provide a formula for the transmission of presence.' Similarly, I am not precisely sure how or when photographers begin to notice things. Seamon (1979, p. 109) himself concludes that 'the unexpectedness of noticing follows no clear pattern. Different people notice different aspects of the same environment' and there is a certain chaos to the whole process. Terence McKenna, in conversation with Abraham, McKenna and Shel-drake (2001, p. 5), notes that 'chaos, creativity and imagination ... each impels and runs the other ... and conserves processes caught up in the phenomenon of being'. What draws the photographer to one subject but not another, what leads them to shoot one moment and ignore the rest, still remains somewhat elusive.

Seamon (1979, p. 106) writes that 'at times watching may be more intense as the interest, beauty or excitement of the scene draws the attention and holds it'. However, like Hahn's dancer, it is a skill that cannot be taught and must instead be learned (transmitted) through practice. In a similar way, the photographers who routinely walk around their everyday environment are noticing, but it is also tightly woven into, and directly informs, their habitual camera use. Noticing is a chaotic, complex contradiction, where everyday life has the potential to be somehow transformed through a balance of luck and skill, and it is a 'synergy between understanding and intuition' (St John 2004, p. 216). Noticing is a state awakened within the photographer; it is a form of rapture that can never be shaken off. One photographer (R1) told me

Eventually, I have become aware about expressing myself: the more you take, the more you look!

Noticing and watching have become a habit and, soon, casual unseen details and the possibilities of everyday life unfurl before the photographer.

The more you take, the more you look, and the more you look, the more you notice. To paraphrase McKenna, it is the felt presence of the camera that makes direct experience of the world heightened for the photographer. From that point, noticing can begin and creativity can blossom every day.

Notes

- 1 The photographers have been anonymized throughout and are referred to by their first name (or user name) initial only. When photographers have the same initial, a number was added to differentiate them.
- 2 This also relates to earlier historical struggles for photography to be accepted as an art form within the academy (see Marien 2002).
- 3 Bourdieu used photographs to link his own concept of 'lived experience' with issues of aesthetics, taste and class (Van House 2011, p. 126). See Bourdieu (1984) for further discussions.
- 4 This was an extension of the walk and talk method. The browse and talk consisted of participant observation of their movements on Flickr, within a variety of locations, with a laptop.

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13 'Analogization'

Reflections on wearable cameras and the changing meaning of images in a digital landscape¹

Paolo Favero

A long way into the digital era we can, today, start looking back at the debate that has defined our engagement with digital images from a critical distance. This field has been characterized by a certain degree of polarization; on one level we have witnessed celebrations of 'digital utopia' (Rosen 2001, p. 318) where the ever-growing number of images circulating on the net, combined with the bare fact that more and more individuals in the world are actively engaging with images, have been interpreted as signs of a move towards a brighter, more democratic future. Indeed, it is easy to get blinded by the mathematics of this change: today, Facebook contains 10,000 times the number of photos that are present in the Library of Congress in the US. Every two minutes, today, more pictures are taken than throughout the entire 1800s (the century in which photography was invented), and so on. On another level, however, the very same developments have also been read from a perspective where concern and nostalgia meet and mingle. Digital images have, in the latter case, been seen as negations of a truer, more direct, 'more real' experience of the world surrounding us, as a proper detachment from everyday life. Gere (2005) has suggested that the move towards the digital generated a fear for 'the annihilation of physical distance and the dissolution of material reality' (p. 15), Wellman (2001) has maintained that the digital has produced instances of 'net-worked individualism' and Nichols (2000) has famously argued 'the chip is pure *surface*, pure *simulation* of thought. Its material surface is its meaning, *without* history, without depth, without aura, affect, or feeling' (p. 104).

The debate surrounding us today is, therefore, much more nuanced. We seem to have overcome the fear of the death of photography postulated by, among others, Richtin (1990) and nicely summed up by Mirzoeff (1999) with his famous sentence '[a]fter a century and a half of recording and memorializing death, photography met its own death some time in the 1980s at the hands of computer imaging' (p. 86). However, we are still caught up in an interesting and unsolved set of doubts regarding digital images' anchorage in the material texture of everyday life, or, to put it in other words, in 'reality'. Such doubts, I will here suggest, largely depend

on the dominance, in the discourse that has characterized the debate on digital practices and technologies, of the notion of digitization and, hence, on the tendency to reproduce (at a discursive and analytical level) the dualism between the material and the virtual.

In this chapter I will attempt to explore these issues by entering the field of digital imaging from an analysis of the changing contemporary scenario of digital technologies at large. In other words, I will foreground the 'digital', identifying leading trends in that context, in order to then look at the specific role played by images in the same scenario. I will open the chapter with a somewhat detailed historical introduction to the core discourses surrounding the birth of the notion of digitization. This passage is needed in order to generate awareness in the reader of the roots (and resilience) of the discourse that has dominated our understanding of digital technologies. In this section, I will also provide a series of glances at contemporary trends in the field of consumer digital technologies. These descriptions will render evident the extent to which we are, today, witnessing the growth of technologies that increasingly close down the gap between technology, the body of the user and the materiality surrounding them. I will suggest that we are in an age characterized not only by digitization but also by what we could call 'analogization' (a process of materialization taking place within a digital environment). I will then proceed to observe the extent to which a similar shift can be detected in the realm of image-making. My analysis of two wearable camera technologies (the 'life-logging' camera Narrative and the action camera GoPro) will lead me to identify the presence of a parallel shift, a 'material turn', which is happening alongside (and probably within) the much-debated 'algorithmic turn' (Uricchio 2011). In the conclusion, I will suggest that such a shift requires from the scholar a capacity to bring the conventional tools of visual culture in contact with elements of digital and material culture and to move away from simplistic dualisms, towards a perspective that integrates the digital with the material (analogue) world. This shift also poses new political and ethical challenges that we must now learn to address.

The digital context

Let me start with a brief historical contextualization. In 1945, Vannevar Bush, an American engineer who had served the US Office of Scientific Research and Development during the Second World War published an article in which he prophesied the birth of a tool capable of functioning like a prosthetic memory (Bush 1945). In times of increasing information flow, Bush suggested, human beings (teachers and scholars in particular) could not be expected to be able to store, categorize, make sense of and use all the information reaching them at an ever-increasing speed. The *Memex* (this is the name of the tool he envisioned) would constitute the solution to this. Built as a memory augmentation system based on microfilm, this was a

desk with translucent screens, an engine and a series of basic commands. The precursor of the modern computer, the *Memex* functioned on the principle of association and was immediately identified as a 'personal' tool in which 'an individual stores his books, records, and communications, and which is mechanized so that it may be consulted with exceeding speed and flexibility' (ibid.).

Bush's intuition did, indeed, cover many of the notions that would influence future IT developments. His writings set the standard for much of the understanding of artificial memory and computers that would come in the following decades. In particular, and this is what I would like to draw the reader's attention to, he formulated a lasting association between electronic technologies and miniaturization (a notion that eventually got mixed up, especially in popular discourse, with that of dematerialization). In his article, for instance, he writes:

The Encyclopaedia [*sic*] Britannica could be reduced to the volume of a matchbox. A library of a million volumes could be compressed into one end of a desk. If the human race has produced since the invention of movable type a total record, in the form of magazines, newspapers, books, tracts, advertising blurbs, correspondence, having a volume corresponding to a billion books, the whole affair, assembled and compressed, could be lugged off in a moving van.

Vanevar Bush's ideas consolidated the notion of digitization that would dominate prevailing understandings of the meaning of digital technologies, up to the present day. His writings clearly identified electronic technologies as tools for making 'virtual' and 'immaterial' that which is actually material and concrete. It anticipated the transposition of books and other paper documents into PDFs, of records into MP3s, films into MP4s, photographs into JPGs, etc.

Popular discourse is, today, to a large extent still influenced by these notions. However, we are already well into an epoch characterized by the opposite movement. A quick look into the market of personal digital technologies reveals that, in fact, we are witnessing a progressive move towards technologies that make possible the translation of visions and abstract ideas into physical items. The 3D printer, to mention one example, is now one of the most popular technologies available in the consumer market. Inverting the principle of digitization, 3D printers provide us with the possibility of giving material shape to abstract ideas. The computer, here, is the vehicle for a process of translation. An idea becomes, first, with the help of 3D design software such as CAD, an image. In the second stage, this image is translated into an STL (Stereolithography) file capable of splitting the image in horizontal layers, hence creating an image/path for each layer. A specific material, called printing ink and looking like a plastic thread but conventionally made up of a variety of different materials (from

recycled thermoplastic to wood, salt, cement, nylon, etc.) is, at this point, liquefied, so that it can be progressively dropped to shape up an image, layer by layer.

Designers, artists and a wide variety of other professionals have recently found a passion for this tool. Since 2013, a 3D Printshow has regularly been hosted in various cities of the world.² More recently, 3D printers have also been adopted in other fields and, for instance, for the production of three-dimensional biological structures, such as patches for heart muscles, livers, etc. With the help of bioink, which is a paste made up of living cells, companies such as Organovo are, today, successfully creating three-dimensional biological structures that may eventually be transplanted into living human beings. Such advancements help us to discover how the cyborg, the futuristic cybernetic organism resulting from the blending of machine and animal/human qualities, object of many fantasies and films (recently portrayed by Matt Damon in the film *Elysium*) is, hence, no longer a futurist projection but very much a reality. We are probably already living in Donna Haraway's (1991) premonition that 'by the late twentieth century, our time, a mythic time, we are all chimeras, theorized and fabricated hybrids of machine and organism. In short, we are cyborgs' (p. 150).

Going back to a more mundane, day-to-day terrain, it is easy to notice how the electronic consumer market is now flooded with products signalling a move towards an increasing interaction with the body. From heavy, bulky terminals to portable computers, we are moving quickly towards mobile screens and wearable electronics. *The Economist* (2014) recently pointed out the boost in popularity of so-called 'smart clothes' at the January 2014 Las Vegas fair of consumer electronics. Smart clothes, that is, garments containing conductive fibres capable of conveying information to and from the body of the users, have recently invaded the market. Mostly tested in the military (where they have been used for monitoring the health of soldiers and also to create flexible circuit boards, connecting all of the soldiers' equipment), these fabrics have also been employed in the world of sport and, recently, fashion, too. In the world of sports, Adidas has recently commercialized a whole line of gear capable of sending information directly from the body of the athlete to the laptop of the coach, thereby helping the latter to monitor the performance of the former. With the help of elastic conductive fibres, the wearer's breathing or muscular activity can be measured, allowing the trainer to monitor the reactions of the wearer's body during a particular exercise. The same materials can also help to gather other biometric data. They can help to signal the arousal of tensions in the body and, so, helping the athlete to identify moments in which to rest or body parts that need particular attention. Adidas Labs, a sub-brand specializing in this sector, has produced shoes tracking everything the foot does, allowing coach and athlete to envision postural problems, etc.

The overall idea of monitoring the body through wearable technologies has, indeed, also been adopted by creators of other types of gadgets. Among them, mention must be made of the health bracelets (such as Fitbit, Jawbone's Up³ and Nike's own Fuelband) which have recently taken the market by storm. Aimed at allowing the user to monitor their own performances (the steps taken on a daily basis, the calories burnt, the heart rate etc.) these bracelets have also inspired the birth of smart watches, which I will return to later. Designers have also recently experimented with the creation of clothes capable of detecting and interacting with the changing mood of the wearer. There are shirts available in the market that change colour according to the user's change in mood, jackets sending out soothing music or messages recorded by loved ones in moments of heightened stress (through loudspeakers built into the collar of a shirt, for instance) and jewellery warning the user of too much exposure to the sun, etc.

A further development of these technologies includes the possibility of not only analysing the data but also sending it back in the form of a stimulus, thereby enhancing performance. With the help of a combination of biosensors and actuators connected to smartphones, tablets or computers, smart fibres cannot only take the user's biometric reading, but also analyse it in real time and then send it back to the user as stimuli capable of modifying performance. Electricfoxy Move⁴ is, for instance, a garment that corrects the posture of the user during Yoga or Pilates exercises. Gentle 'taps' help them to recognize incorrect postures and suggest how to adjust a movement while it is being executed. Intelligent fibres are also being tested which generate small electrical circuits, activating or deactivating specific muscles.

There is no need to go any further into this for the moment. The products mentioned so far are sufficient to demonstrate the variety of bodily and material engagements realized with the help of digital technologies. However, in order to conclude this section I would like to discuss briefly the launch of Apple's Watch, an event that has, somehow, sanctified this trend.

Apple's recent launch of 'Watch' constitutes the epitome (both in technical, as well as in metaphorical terms) of the process of progressive engagement with the body and with everyday life. The aim of the Apple Watch is, in fact, to engage a continuous dialogue with the body of the user. Meant to be worn all the time (showers included), it synchronizes, thanks to the sensors placed underneath the watch and the band, with the body of the user, constantly gathering information about its functioning. The Watch (which is also armed with an internal GPS tracker) communicates with the iPhone, iPad and the computer, creating a circuit of information that, from the point of view of the user, can be monitored through a series of applications (particularly the 'health application'). These apps, together, collect information regarding the users' eating and sleeping

patterns, their biometrics, their fitness habits, their commuting routes, working and relaxation practices, etc.

In line with some of the technologies discussed above, the Watch also uses tactility in order to communicate with the user. Small haptic signals (read 'taps') on the wrist inform the user that it is time to get up, that an SMS has arrived etc. The Watch promises to be the ultimate tool for allowing users to gain control over their own bodies, living habits and health patterns. However, at the same time, it also constitutes the ultimate opportunity for creating the largest database of biometrical data, behavioural and living patterns, shopping and leisure habits, and for putting that data into the hands of market interests.⁵ The Watch seems to close a circuit between the body of the user, their phones and computers, and the market⁶ in a movement that demands heightened attention from the point of view of both ethics and law. It is not my aim to enter this debate here. Instead, I want to stress how the gradual diminishing of the distance between technology and the body can also be detected in the marketing of the Watch. The leading buzzwords and slogans are 'Our most *personal* device yet', 'A more immediate, *intimate* way to connect', 'intelligent health and fitness *companion*', 'because it *touches* your skin, we were able to add a physical dimension to alerts and notifications' (my italics). We could say that Apple is actively tapping into the trend towards increasing bodily engagements, somewhat sanctioning their present centrality in the context of digital technologies.

The examples cited so far should suffice to show the vast span of bodily engagements that contemporary digital technologies are making possible. Basically, there is a whole market of digital products out there that are directly engaging with our bodies and with the material texture of everyday life, rather than just 'simulating reality'. Such examples may have also shown how technologies and practices that we may still have a tendency to consider avant-garde and futuristic are, today, actually readily available 'out there'. They are affordable for a growing number of citizens who live in the networked parts of the world and who are armed, of course, with the necessary capital.

The visual context

In the field of image-making technologies, there are many examples of the present shift towards an increasing anchorage of images and of the practice of image-making within the materiality, physicality, spatiality and sociality of everyday life. The most obvious example is perhaps that of Google Glass, the first case of wearable vision-based augmented reality technology. The Google Glass allows users to receive 'virtual' data on the lens (thereby overlapping with the direct perception of the world 'out there'), that is, a similar kind of information to that which we conventionally receive on a smartphone screen (such as directions on a map, information about places

to visit, incoming calls and messages, etc.). It brings together geolocate (spatial) and geosocial (relational) information in the field of vision, constituting a brilliant example of how images now force us to go 'beyond the frame', as I have suggested in a recent article (Favero 2014). Capable of attracting the attention of critics and scholars from all over the world the Goggle Glass has, however, failed to attract buyers and was recently removed from the market.⁷ A recent article in *Time* (Grossman & Vella 2014) which, by the way, is openly flirting with Apple products, suggests that the reason for its failure can be found in its invasive look, which demands attention from bystanders and users alike and therefore interrupts genuine immersion in everyday life.

On a different terrain, an increasing attempt to blur the distance between the image and the sensorial richness of our immediate perceptions can be detected in the growing popularity of multi-sensory cinema halls (the so-called 4D, 5D or even 7D movie theatres). Here, the act of viewing is accompanied by sound, touch and motion related stimuli and, in some cases, also by the use of smells and temperature changes. Also, connecting back to the aforementioned discussion on 3D printing, we could take into consideration the enduring popularity of 3D portraits. Companies like Twinkind, Omote 3D, MiniMe, etc., have recently started promoting the large-scale production of 3D portraits. With the help of a photogrammetry 3D scanner, a technician takes a full 360-degree image of a specific subject. This image is then converted, with the help of a printer, into a small puppet representing the subject. The commercial (and, at times, also technical) fortune of such technologies has indeed been characterized by ups and downs and, for the moment, they seem mostly to constitute gimmicks. However, they contain the seeds of some of the transformations that I am addressing in this chapter, i.e. greater immersion of viewers in images, and of images in the fabric of everyday life.

Such practices constitute a challenge to our conventional way of addressing, for instance, a portrait or visual representation more generically. In the context of, for instance, 3D portraits, we lose the conventional, privileged point of observation that photography inherited from Renaissance practices of lifelike depiction (see McQuire 1998; Crary 1990). Let me, however, move on to a couple of more specific examples that will direct our attention not only to the use and consumption of images but to their very production. I will proceed to analyse two popular technologies: one life-logging camera (the Narrative Clip) and one action camera (the GoPro). I will begin with the former.

The Narrative Clip: time, mimesis and monitoring

The Narrative Clip (formerly known as Memoto) is a Swedish-made, small, lightweight, wearable and fully weatherproof camera. Armed with a clip, allowing it to be pinned on a jacket/bicycle, etc., it takes two shots

every minute. Such shots are recomposed, with the help of a narrative algorithm, in a time-lapse sequence that the user can view through a dedicated smartphone application. These sequences are divided in discrete blocks (labelled 'moments') that also display GPS and temporal data. The marketing material for Narrative promises the user that they will be able to track specific events happening during the day and then use it, in a McLuhan and Fiore (1967) fashion, as an extension of their bodies and minds. A prosthetic memory, capable of making up for what our mind is not able to do, Narrative produces a kind of visual diary of our day. Devoid of a viewfinder, this camera does, indeed, pose a number of interesting challenges to photography. Primarily, it dissociates the act of image-making from the intentions of the image-maker, that is, from the processes of selection and interpretation that are an intrinsic part of the act of making pictures (choosing the right angle, framing, selecting the moment, etc.). It therefore constitutes a movement towards a somewhat 'purer' form of mimesis.⁸

I must admit that using a life-logging camera is quite a bizarre experience. My main concern, when experimenting with the camera, was really the safety and physics of its positioning. I was concerned about having placed it in the right position and about not losing it. Thus, my expectation of total mimesis and spontaneity was constantly interrupted by such practical concerns. With time, this aspect did, however, improve and I learnt to experiment with different positions. On one occasion, I even ended up forgetting the camera in the lower corner of my coat. Surprised and relieved by having found it, I still had to acknowledge that the material produced from this angle was quite useless.

The material the camera produces requires a set of specific reflections. We are not talking, here, about something different from what the marketing of the product says about good photographs, or just photographs for that matter. Rather, what we bring home with Narrative is, largely, a set of scattered visual impressions (most of the images are underexposed or blurred). With the help of the online mobile phone application, however, the user can choose to view only the images with good light and definition. This, however, brings us to the first problematic aspect of this camera. In order to exploit fully its potential, this camera requires an online subscription (of US\$10 per month), which henceforth binds, similarly to the Apple Watch, the user to the market. Detached from that subscription, the user may not access GPS information or share images on social media, etc.

Inserting the act of making images into everyday life, in these somewhat novel ways, life-logging cameras raise a number of ethical and legal concerns. The people in front of the user may not be aware of being photographed by that little tool which looks, more or less, like a tag or a small iPod. On one occasion, for instance, I was able to walk across a metal detector in an airport with my camera 'capturing' the whole process. So, in what way does it distinguish itself from a CCTV camera (for which there

are clear principles and laws)? From this point of view, Narrative resembles other technological items, such as the cameras which can be placed on the dashboard of a car and record everything that happens in the traffic and, at times, also the actions of the driver.

I will get back to the political and ethical implications of these technologies in the conclusions. For the moment, let me instead point out the extent to which the features of the Narrative Clip that I have described so far resonate with some of the broader conceptual transformations that have been debated in the context of contemporary image-making practices. In the first place, this camera seems to materialize the instances of bifocality that Peters (1997) described, some time ago, in terms of ongoing dialogues between the local and the global. According to Peters, our experience of the world, today, is constantly mediated by technologies of representation (he uses the examples of the satellite maps used in weather forecasts). Such representations make us constantly aware of who we are in relation to the context in which we live. A phenomenon that is presently embodied by the various forms of emplacement that the integration of image technologies with GPS positioning have made possible (cf. Lapenta 2011; Pink & Hjorth 2012), this aspect is present in the very functioning of the Narrative Clip, which transforms the banality of a regular day at work into an ongoing process of representation. However, pushing Peters' insight further, the Narrative Clip shows us that, in the present age, we are not only constantly being represented but also capable of representing. In fact, while CCTV cameras, satellite imaging and weather forecasts have made us aware of the fact that we are the object of constant monitoring, mobile phone cameras (and ultimately life-logging cameras) stimulate our agency as producers of images, our capacity to 'picture' (Heidegger 1977) and, hence, potentially to dominate the world surrounding us (cf. McQuire 1998).⁹ Resonating with one of the fundamental insights on vision (described by, among others, Berger 1972 and Barthes 1977 but also by phenomenological inspired scholars such as Pinney 2001), Narrative reminds us that vision is a reciprocal, dialogical fact and that the act of seeing always corresponds to that of being seen. The Narrative Clip is one of the tools now making us realize that we are, ourselves, part of the visible world.

GoPro: materiality and the opening of the image to the senses

Let me now switch to another technology that embodies a different form of engagement with the mundane. I must warn the viewer that I have now to briefly shift away from photography proper and into that blend of photography and video that characterizes much of image-making in a digital landscape. Action cameras were launched in the world of extreme sports and were originally designed to allow the production of images in

conditions and places that would not permit the use of regular cameras. The famous GoPro, for instance, one of the leading companies in the sector and the camera that I will use here to exemplify my reflections, was designed by founder Nick Woodman, in 2002, in order to film the practice of surfing from the point of view of the surfers themselves. It responded to the need to immerse (metaphorically and physically) the camera in a place and a moment that could not be captured otherwise, at least not in the absence of impressive technological and infrastructural investments. Technically, the GoPro produces HD videos and photographs in different formats. It mounts a fixed lens, capable of capturing a 170-degree angle, and can be paired up with another similar device in order to produce 3D images as well. Mounting a shield, protecting it from water (to a depth of 40 metres), scratches and bumps, GoPro cameras can be mounted on the body of the user with the help of straps, thus generating, similarly to the Narrative Clip, an interesting intimacy between body and device.

From being a niche product, GoPro (along with other similar cameras) quickly took the market by storm and became a proper consumer item. Costing between €250 and €450, it is, in fact, available to a quite large proportion of the population today. Combined with its easy functioning (similarly to the Narrative Clip, the user does not need to be a professional) it contributes to the popularization of filming possibilities that were, once upon a time, available only to a limited crowd of specialists. Recently, professional film-makers have increasingly adopted the camera, too, and, among them, Harvard-based ethnographic film-making duo Lucien Taylor and Verena Paravel. Their multiple award-winning documentary *Leviathan* offers a sensorial exploration of the world of open sea fishing. We are offered images of fishermen enduring the hardship of life on the boat, along with an ongoing exploration of the natural world surrounding them. Fishes and men, it could be said, are both equally at the centre of this film and this has been made possible thanks to the use of the GoPro. Taylor and Paravel creatively engaged with the camera (actually, as I was informed by members of the sensory lab, with a large number of GoPro cameras, many of which were lost during filming), inserting it in a variety of different situations. They mounted it on to the bodies of some of the fishermen, thus allowing observation of a set of delicate and potentially dangerous operations that would hardly have allowed for the presence of a cameraperson. They also mounted it on various bars on the boat, to offer us glimpses of the sea and of that thin line separating the sea and the water. In this way, we are offered the opportunity to gaze at the sea, with its wondrous living creatures and, during fishery, also animal debris. In some shots, we see the boat from the perspective of the sea and are invited to, in a way, immerse ourselves in the blood resulting from the first process of cleaning and selecting the fishes. We also follow the fishes from close quarters, as they go through this process of selection, and have some magnificent shots of seagulls following the nets in the hope of finding some

food, as well as 'wet', 'hazy' images of the fishermen in the showers. David MacDougall suggested that there is something 'cosmic' about this film.¹⁰

Basically, we may conclude that the GoPro camera supported Taylor and Paravel in their in-depth, sensorial exploration of open sea fishing. It allowed them to offer a unique window on to a world otherwise out of reach for most human beings (thereby living up to the broader objective of documentary film-making in general). What appealed to me, during a first viewing of *Leviathan*, was the extent to which these cameras allowed the generation of images which went beyond conventions of movement in documentary film-making. Lying there, amidst the fishes or in the sea, the camera is not guided by the intentions of the film-maker but rather by its various engagements with the materiality of the elements surrounding it. This, I also experimented with personally, during my first engagements with this camera. As for most other users, I guess, I felt, right after purchase, immediately compelled to throw the camera straight into the water and experience, by letting it hang with a cord, its way of engaging with water, with sea weed, etc. In *Leviathan*, interesting movements are generated when the camera is mounted on to the fishermen. Here, we are offered images that move according to the needs of the fishermen and not the conventions of cinema (there is, for instance, an overload of canted images in this film).

We are, in other words, offered a series of 'unconventional' yet very revealing point-of-view shots, centring on the fishermen (and also on fishes, the water and on the boat itself). In these moments, the camera functions like an opening, a window on to the other senses. We experience the frailty of balance on a boat oscillating between the waves, the anxiety of the fishermen reflected in their breathing, the sounds that they produce and receive from their own vantage point. We are offered a quite unique sensorial experience of life on a ship. Interestingly enough, the GoPro captured my attention for its capacity to re-produce sounds that are, once again, hardly ever captured with 'prosumer' technologies. This is true for the underwater scenes in *Leviathan*, where our attention is caught by the sound of the water, the squishy noise of the fishes, etc. This was also achieved through my own experimentation with the camera, in an entirely different type of context. I have, in fact, used GoPro cameras in the context on a project focusing on a tennis coach and his teaching methods (see Figure 13.1).

For this project, I mounted cameras on top of players' heads or on their chests whilst they trained. This allowed me to gain a particular perspective on the choices of timing that they make on court (how and when they move in relation to an arriving ball, etc.) and portray one specific activity from the point of view of the subject performing it, that would be difficult to capture in the absence of these cameras. What caught my attention the most, however, was the players' breathing patterns. I have noticed when they inhale and exhale, and when and how they use their voice (shouting) for coordinating their game, etc. The proximity of the camera to the body of the user thus highlights the coming together of the senses, the way in



Figure 13.1 Still from a videoclip taken with a GoPro camera mounted on the head of a player (source: photograph by Paolo Favero, I acknowledge the support of coach Giampaolo Coppo who let me involve his players in my experiments).

which their act of viewing the ball merges with that of touching and feeling it and also with the sounds produced by the body. In the context of a sport such as tennis, GoPro cameras are, in fact, still too bulky and interrupt the spontaneous flow of the body. Nonetheless, they open up a space of exploration, through audio-visual means, that was previously not available to us. In the context of my own work, as in *Leviathan*, the GoPro camera has been functional for opening (to paraphrase MacDougall) a 'pathway to the other senses' (MacDougall 1997, pp. 289). As a tool, it allows us to engage with the materiality of life amidst the most diverse range of situations, closing, by melding with the body of the user, the distance between the image and the sensorial texture of everyday life.

Let me point out that today there are many other tools that offer an opportunity for a closer sensorial exploration of the things surrounding us. Macro and fish eye lenses that can be mounted on top of our smartphones can now be purchased at very cheap prices and reveal to us fascinating aspects that escape the human eye and conventional cameras too. This is the case with the photograph in Figure 13.2, which I took with my iPhone and the help of a mobile cam macro lens.

Conclusions

The discussion above has shown how the Narrative Clip and the GoPro camera constitute two different modalities through which contemporary



Figure 13.2 Photograph of a snail taken with a macro lens mounted on an iPhone 5 (source: photograph by Paolo Favero).

image-making technologies enter the texture of everyday life. I suggest that such technologies (and the practices associated with them) show the extent to which contemporary image-making digital technologies are characterized by what we could call a 'material turn', that is happening alongside (or, rather, within) what Uricchio (2011) called the 'algorithmic turn'. While the latter (which is more closely connected to the notions of digitization that I mentioned at the beginning of this chapter) seems to have largely dominated the debate on digital imaging, in the present day we must acknowledge the co-existence of this material shift. This is no longer the age of digitization but that of analogization, of a process of increasing attention and production of material instances that is happening within (and not in opposition to) a digital habitat. Parallel to the variety of algorithm-driven virtual reproductions of the 'real' (see, for instance, the much debated Photosynth software or the Oculus Rift immersive virtual reality goggles), we are now witnessing the 'real' reproduction of the virtual.

Introducing the act of image-making in a context defined by notions of wearability and bodily proximity, the cameras that I have discussed here seem to ask us to suspend some of the conventions that characterize our

ways of engaging with (and understanding) the act of making pictures. Devoid of viewfinders, they cut off the intentionalities of the film-maker. Foregrounding mimesis rather than interpretation, they diminish the importance of notions of proper framing, composition, timing, etc., that is, the conventions associated with photography and film-making. This decen-tring can result, at times, in a creative process offering the image-maker new unexpected perspectives. In fact, the user does not know exactly what they have 'captured' with this camera until we import the images to the laptop. As one film-maker using GoPro told me, 'it feels a bit like going back to the past: you film and only at night you discover what you have actually done'.

In the context of both the GoPro and the Narrative Clip, it is the body of the user that functions as a viewfinder, determining what will be contained and what will be excluded by the image. The body is literally the eye of the camera. In a paradoxical way, rather than functioning as prostheses of our bodies, Narrative Clip and GoPro seem to ask our bodies to become prostheses of the camera, thus inverting the process that MacLuhan anticipated decades ago. The manner in which these cameras seem to close down the distance between the image and the world surrounding it (which depends upon a combination of wearability and the use of short lenses) also seems to resonate well with ongoing trends in online digital photography, which are characterized by the increasing portrayal of domesticity, of whatever is in close proximity to the user. Instagram, for instance, is a gallery of coffee mugs, feet and half-eaten dinner plates, etc. As Murray (2008, p. 151) has suggested, photography has started dealing with 'an immediate, rather fleeting display of one's discovery of the small and mundane (such as bottles, cupcakes, trees, debris, and architectural elements'.

There are some major differences between the Narrative Clip and the GoPro. The particularity of the former is that it inserts, through the principle of the time-lapse on which its functioning is centred, the act of picture-taking into everyday life, as a somewhat random activity defined by time. This, too, mirrors another central feature of contemporary digital photography, which can be exemplified by the sharing and viewing practices which characterize common usage of Flickr, Facebook, Twitter and Instagram. Here, photography, rather than appearing as a technology for preserving time or, to use Bazin's (1967) term, for 'embalming' it, serves the purpose of creating a shared simultaneous experience of time, for the creation of what Ito long ago called a 'visual co-presence' (2005). This shift sums up what MacQuire suggested when he wrote that today '[p]hotography is becoming less about capturing "memories" (as Kodak famously phrased it in the 20th century) than about commenting on present events as they are taking place' (McQuire 2013, pp. 226). Similarly to Instagram, Facebook, etc., Narrative Clip foregrounds time as the conceptual centre on which our engagement with images is centred. This

happens both at the level of production and of viewing. Besides taking pictures at regular time intervals, the Narrative Clip also forces the viewer to explore the images taken through a time-defined structure. The images are, in the first place, viewed in the form of a video clip made of individual stills.¹¹ Second, such moments can be posted, similarly to Instagram, Facebook, etc., on the basis of a timeline (with the newest elements on top and the oldest at the bottom), thus foregrounding their 'ephemerality', to use Murray's (2008) term.

The scenario opened up by the GoPro is, instead, characterized by issues of sensoriality and by a playful entanglement between the camera and the material environment surrounding it. The GoPro, as demonstrated through the examples of *Leviathan* and of my own experimentation on the tennis court, allows the image-maker to capture sensorial information that centres on the body of the subjects filmed. Hence, it enters places that were previously out of reach, making accessible to amateur viewers and image-makers alike, a world that was previously out of sight. Through the GoPro (and also the macro lenses that I described above), the act of image-making seems to be given back that magical, transcendental connotation that, according to Benjamin (1931) was at the core of the history of photography. According to him, photographers were, in fact, the descendants of the augurs and haruspices, and the camera a tool for entering the 'optical unconscious', opening up a new perception of the environment surrounding us.¹²

As these lines may have suggested, the technologies I have addressed in this chapter do not only constitute an opportunity to glance towards the future but also to rethink critically the past and to reinsert it in our present analysis. Rather than pulling us out of the history of photography (and film-making) and into an unknown terrain, contemporary technologies seem to bring the notions and debates regarding the meaning of photographs that have lain at the centre of the study of visual culture, back to our attention. I have also shown how such a re-evaluation requires a capacity to bring the analytical tools belonging to this field in touch with those belonging to material and digital culture.

Let me now go back to where I started from, that is, from the world of digital culture. The progressive engagement with the body and with the material texture of everyday life that I have identified in this chapter is mirrored in the cultural history of cybernetics. Hayles (2010, p. 148) has, for instance, suggested how 'Instead of constructing virtual reality as a sphere separate from the real world, today's media have tended to move out of the box and overlay virtual information and functionalities onto physical locations and actual objects.' The present age, she suggests, is characterized by the coming together of the animal, the machine and the human (the fish eye shots of fish eyes and humans contained in *Leviathan* nicely symbolize this). So, as this chapter has hopefully made clear, this is no longer the era of virtualization but one in which the material and the virtual meet and

minge, giving birth to new (at times curious) tools and practices. Besides posing interesting theoretical challenges, this progressive confluence is, however, also raising a series of crucial political and ethical questions. The desire to constantly 'picture' (Heidegger 1977) and explore, in greater detail, the world surrounding us does seem to translate itself, paradoxically, not just into an exercise of freedom (containing the seeds of a future cyber democracy) but also into an act of monitoring the world in which we live.

The disquieting question we must ask is: on behalf of whom is this monitoring enacted? MacLuhan ([1964]/1994) anticipated this question long ago when, commenting upon audio-visual media, he feared that our eyes, ears, hands, etc., no longer belonged to the bodies they were attached to but rather to the multinational companies they are connected to. This is what the Apple Watch seems to represent: a window on to a possibly post-post-modern, decentred Panopticon (or Plenopticon).

In *Discipline and Punish*, Foucault (1977) showed us how, in the passage from Renaissance to modernity, the relation between vision and power had changed significantly. In pre-modern times, the fact of 'being looked at, observed, described in detail ... was a privilege' (p. 191), he suggested. This is nicely embodied by the Hall of Mirrors at Versailles or by the tradition of royal parades (i.e. those moments in which the King would give the people the right to look at him, exercising, simultaneously, his right to be looked at).

In the Modern Age, however, the hierarchy of this relationship was reversed. The act of looking became synonymous with control, monitoring and, consequently, with discipline. This is what Bentham's Panopticon was about. An innovative, more functional and humane way of building a jail, this was a model for disciplining society through vision. In this context, the King no longer had to be looked at but was rather the subject who looked at others. Metaphorically seated at the centre of the Panopticon, he progressively transfigured himself into a kind of Big Brother, seen by no one but able to see everyone. Applying this metaphor to the contemporary context, we may wonder where the King is located and what rights and obligations he has today. No longer the object of a privileged gaze or, himself, an active observer, he (or she? or it?) may, perhaps, be gently leaning back on a throne of big data, relaxing as we monitor and discipline each other in our ongoing playful orgy of visual co-presence.

Notes

- 1 Research for this chapter has been made possible thanks to the generous grant awarded by the Science Foundation – Flanders (FWO) for the project 'Defying Social Invisibility: A Transnational Study of Empowerment and Social Intervention in the Field of Contemporary Digital Imaging Practices' hosted by the Visual and Digital Cultures Research Center (ViDi), University of Antwerp, Belgium.
- 2 <http://3dprintshow.com/home-2014/>, viewed 10 January 2015.

- 3 Mention should also be made of the way in which US-based company Jawbone used their fitness tracker, a bracelet pedometer/sleep-tracking device, to study the behaviour of people living in the area struck by an earthquake of magnitude 6.0 in Northern California on the 24 August 2014.
- 4 www.electricfoxy.com/move/, viewed 20 October 2015.
- 5 This is emblemized by the newly designed payment method that can be easily accessed through the Watch itself.
- 6 I cannot resist pointing out how the name 'Apple Watch' contains a kind of omen of its desire to control the user, rather than letting them control it.
- 7 At the time this chapter is going to press, news is spreading that the Google Glass may soon be back on the market again.
- 8 Foregrounding the habitual movements of the body of the user, as a kind of viewfinder, it also constitutes the final form of disentanglement of the photographer's gaze and that of the camera, completing the transformation that was started by the introduction of LCD screens on digital cameras.
- 9 As McQuire (1998) suggests, when we photograph from afar we get used to the idea that not only can we take possession of that far away view but, also, that someone else from a far away position can view us and take possession of us through their own view.
- 10 Personal communication.
- 11 The viewing of images as moving-image blurs the boundary between photography and video, which is another key characteristic of digital imaging (see Favero 2013).
- 12 Pinney summed this question up by saying that technology 'suggests the apparatus of the camera and its chemical way of referring to the world. Magic suggests a contagion of qualities and the ability to produce effects beyond the range of ordinary bodies' (Pinney 2001, p. 12).

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16 Is the camera an extension of the photographer?

Martin Lister

I wish to offer a reading of an image, if only to leave it behind. My reason for doing this – both the ‘reading’ and the ‘leaving behind’ is to engage with the idea contained in the title of this part of the book: that the camera can be thought of as an extension of the photographer. It also allows me to mark a current shift in how we are thinking about photography: the turn away from a long dominant preoccupation with photography as representation. The image is Andreas Feininger’s 1951 photograph entitled ‘The Photojournalist’. It is an image that invites thought about the mechanical extension of the human while hinting at the limits of the idea.

The non-representational turn

The act of finding meaning in a photograph is, of course, to engage with photography as representation. This, in turn (if it is not to be an innocent reading), inevitably entails a measure of academic discipline and methodology: the semiological scrutiny of images treated as texts, with the aim of revealing or interpreting the meanings encoded within them. With regard to photography, this is a developed practice that, over the last 30 years or so, became almost synonymous with photography theory.¹ Now, the ongoing convergence of photography with computing and the rapid development of photography as a networked and computational medium have rendered photography radically more transient, relational, dynamic and polymorphous; we have witnessed a kind of supercharging of what it already was! The sheer degree of this change has rendered the analysis of singular images as discrete artefacts as largely inappropriate; the object of theory has changed.

This transformation in photography has been accompanied by a turn, in theory, to what has been called ‘non-representational’ or, better (if less common), ‘more than representational’ theory (Lorimer 2005, p. 84). This is a term that points to a diverse, unsettled (and deliberately unsettling) cluster of approaches to the study of social life and cultural practice, which has been developing for quite some time.² Its main home is in the discourse of human geography, but it has strong resonances in science and technology studies, particularly ‘actor network theory’ (Latour 2005).

The goal of non-representational theory is to shift attention away from the decoding and revelation of the meanings that we take to inhere in representations, and towards the noise, habit, impulse, emotion and unruly activity that surrounds and exceeds them. For the non-representationalist, the world understood as meaningful text, as representation, is a conservative and restricted view and is replaced by a sense of its material and phenomenological richness. Such approaches are interested, therefore, in the embodied and material practices of human social actors and their traffic with non-human things and the physical environments in which they move and meet. Where photography is concerned, this is a challenging project. It asks us to shift our attention away from its products – photographs – and the attention we have traditionally paid to them, to its practices or, as non-representational theorists like to say, to its ‘doings’ (Larsen 2008, p. 146).

Yet, ‘turns’ in theory come and go and visual representations do (continue to) tell us things about the world. Radical shifts in photography and photographs, on many levels, do not mean that, with respect to some photographs at least, reading an image as a text cannot still be useful, cannot be ‘telling’. For all the important emphases that ‘non-representational theory’ foregrounds, to eschew representations completely would surely be a bad case of throwing babies out with the bathwater. Andreas Feininger’s iconic image, made a little over 60 years ago, is an example in point.

Camera as tool

In Feininger’s image we see a Leica camera held close to the photographer’s face. The camera’s lens and viewfinder are aligned, horizontally, to the photographer’s eyes, its body parallels his nose, as his fingers curl around its body. The camera is anthropomorphized; it is depicted as an extension, almost a part, of the photographer’s face and head. This is an image that might have been made to illustrate the title of this part of this book.

Feininger’s image offers us a certain concept of technology: the anthropological concept of a tool. The camera is pictured as a hand-tool that extends the photographer’s body and its sensorium, much in the way that a hammer, a screwdriver or a knife, are seen to extend the body. Such extensions are then held to amplify and refine the body’s capacities and, in doing so, they facilitate a human subject’s power and intention. Understood in this way, tools are wielded, they are under the control and direction of human subjects. Throughout the twentieth century, the photographic camera, along with the microscope and the telescope, was a paradigm for understanding technologies as extensions of the body, in particular, as extensions of the human eye.

In the era of analogue photography, photographers *used* their cameras. They controlled the work that they did. At the professional and serious



Figure 16.1 Andreas Feininger, 'The Photojournalist', 1951 (source: photo by Andreas Feininger/The LIFE Picture Collection/Getty Images).

amateur end of the spectrum of practice, this was achieved through a mastery and coordination of a number of factors: principally focus, shutter speed, aperture and depth of field. Further, the photographer's choice of film type (what we can now understand, retrospectively, as the analogue camera's 'software') and the manner of its chemical development and printing, gave control of other factors such as tonal contrast, grain, detail and colour. At the level of consumer and snapshot photography, these processes were simplified and automated via a set of simple controls, whose functions were usually announced by small icons on the camera's body. Here, as has been frequently observed, commenced the 'black boxing' and rendering invisible of photographic technology, a process that

has developed exponentially as cameras have become software machines (Palmer 2013a, p. 49ff.; Slater 1995, p. 129ff.; Stallabrass 1996, p. 13ff.; Rubinstein & Sluis 2013, p. 25ff.).

In the manner of tools, analogue cameras were discrete and bounded objects. However finely crafted or mechanically ingenious, when not in use, they were inert things. While some auxiliary items could be added to such cameras (for example, a flashlight, a light meter, a motor drive), cameras were discontinuous with other parts of photographic practice or industry, its means of production, storage, distribution and reception. In more concrete terms, the analogue camera was a tool for making a photographic exposure (note to Polaroid) and it required other, often remote, places, institutions and apparatuses for a photograph to be produced: the chemical darkroom and enlarger (where the latent image became visible) and the physical carriers and means of distribution, circulation, display and storage (magazines, newspapers, albums, archives and transport systems of various kinds). While it was already common, from at least the mid-twentieth century onwards, to think of a photograph as a captured 'moment' in time, as something produced in seconds or fractions of seconds, this applied only to the moment of exposure. The full temporal and spatial extent of the analogue photographic process could be considerable, entailing dedicated buildings (darkrooms, laboratories, print shops) and the time and means to travel between these places, across which the division of photographic labour was distributed (Gómez Cruz & Meyer 2012).

However, when in use, the analogue camera was far from inert: it shaped how the photographer looked at the world and how they were positioned to look. Returning to Feininger's photograph, we see that the photographer is looking through a viewfinder that is attached to his Leica camera. Here, human vision is channelled or funnelled and the otherwise mobile and saccadic field of view of the human eye is framed and fixed, it is restricted by the hard metallic edges of the viewfinder's window and it is supplied with a perspectival image constructed by the lens. By these means, the use of a viewfinder also established a distance between the viewer and what or whom they looked at. Scopic power is involved here, as the distance is not only perspectival and optical, it is also physical and psychological. The photographer is set back 'behind the camera' and apart from the world they photograph. A corollary of this distance is the licence or presumed authority to look or gaze at others in an unflinching manner that is hardly possible with the unaided eye (Lutz & Collins 1991).

The networked computational camera

When considering contemporary forms of the camera (the digital camera and the camera phone), Feininger's image of the mid-twentieth century photojournalist ceases to be so instructive. Photographers now view the

world on the LCD screens that occupy the full width and height of the camera's slim body, held, with extended arms, away from the photographer's body. As they sweep and scan the environment (the pro-filmic space) the digital photographer watches the world becoming an image, or one image after another, on the camera's mobile screen (see Forrest, this volume). Rather than being distanced or placed outside them, it is likely that the digital photographer will be close to the centre of the events and activities they photograph (see Favero, this volume). Where the viewfinder once conferred a privileged separation on the photographer, the digital screen is likely to prompt a fair degree of social exchange between photographer and photographed, which takes place around the very act of photographing. Moreover, on the same screen, the photographer, and those photographed, can view an almost instantaneous 'playback' of whichever image was chosen. In short, the choosing or conception, the exposure, the storing, and the display and initial reception of an image, all take place within the digital camera. With internet connectivity and the downloading of photographic 'apps' (Palmer 2013b, pp. 157–159) the distribution and, to a degree, the processing of the photographic image can be added to this list.

These capacities of the digital camera mean that it is no longer a discrete tool, passively awaiting its utilization by a photographer. The camera has become a computer – a computer with sensors – and an element, if a key one, within an extended system of connected technologies (see Lehmuskallio, this volume). Photography has become algorithmic and computational (Rubinstein & Sluis 2013, pp. 21–40), and the computer and network technologies of the twenty-first century that now constitute 'digital photography' are no longer discrete, monolithic machines and dedicated places, they are disaggregated and distributed. They pervade our environment, they are ubiquitous, they are 'everyware' (Kember 2013, pp. 56–76). Digital technologies, generally, have become pervasive and embedded in the physical environment we inhabit.

Contemporary cameras, whether in the form of the camera phone or as freestanding digital cameras with wireless connections, exist as devices within a communications network. This network is always active, it is always 'on'. As Rubinstein and Sluis put it, the network observes no holidays or weekends; it exists in a continual present (2013, p. 31). The images that enter this network via such cameras do not follow a linear path: they can be in many places and contexts simultaneously. The singular capture of a latent image in the old 'camera tool', and its subsequent path through the dedicated places and institutions of materialization, distribution and reception, has given way to a process driven and managed online by software. Further, such software does not only dwell within the new cameras themselves, but also within the other computers and online sites to which they are connected and where images are stored, organized and processed.

Edgar Gómez Cruz has argued that we should not understand photography as representation, or solely as technology or object. Rather, photography needs to be understood as the outcome of a complex and changing alignment of technologies, meanings, uses, knowledges, institutions and practices (Gómez Cruz & Meyer 2012, pp. 204, 217; Gómez Cruz, this volume). This is a way to understand photography, in its historical forms as well as in its current condition, as the alignment between this field of actors and agents changes over time. Within this field, the notion that one agent might simply extend another no longer holds. In his thoughts on the current relationship of cameras to software, Daniel Palmer suggests that the 'creative act (of photography) no longer belongs to the photographer alone, if it ever did, but is deferred to software and to increasingly collaborative possibilities (both human and non-human)', and this means that 'the traditional role of individual human agency in photography is changing' (2013b, p. 63).

Maybe a final return to Feininger's photograph is called for, after all, in order to conclude this brief reflection on the changed nature of the camera. In the light of the observations made above, that is, when thinking about digital networked photography, the cyborgian play in Feininger's image, between a body and a machine, between a photographer and a camera, begins to imply something less clear cut and one-directional than my initial reading suggested. It prompts us to invert our title and to consider that the photographer has now become an actor positioned within a dispersed field of non-human agents, to which intelligent and networked cameras give access. In this light, and if the concept of 'extension' is to continue to serve us, we might need to say that, rather than the camera extending the photographer, the photographer has become an extension of the camera.

Notes

- 1 See, for instance, Burgin (1982), where he influentially argued that a theory of photography should be, mainly, a theory of representation.
- 2 Even within cultural geography, not to mention its other versions in science and technology studies and philosophy, a bibliography on non-representational theory would be very extensive. However, portals to the field and useful critical and historical reports on non-representational theory are to be found in Haydn (2005 and 2008).

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17 Outlook

Photographic wayfaring, now and to come

Nancy Van House

This book argues for a practice-theory-based approach to everyday photography, emphasizing people's concrete activities and understandings around photographs, photography and photographic technologies, and how photographs are meaningful in people's social lives. This book is particularly concerned with networked camera devices and the connected interactions they support.¹ This is a good time for a book like this, since technologies-in-the-making are particularly useful for making visible what soon comes to be invisible and taken-for-granted (Latour 2005).

In this chapter, I explore some of the current developments that, I believe, are fundamental to people's experiences of photography and its entanglement with ways of being in the world. Participation in photography has become more extensive and diverse, bringing in new people and transforming practices, especially in association with social media. Images are, in effect, more personal and collective, as well as less durable and more immediate, altering time and place. As images become more public, people don't just see but experience others' points of view. Photos remain important to memory, self-presentation and the enactment of identity, but with important differences. In the second section, I briefly address developments in theory and methods. In the final section I ask what all this means for photography in people's lives and why images are so important.

Significant developments, and a caveat or two

One of the most important current developments associated with networked, digital photography is a massive increase in participation. Participants are critical to practice: without practitioners, there is no practice. Practices and participants change one another. Practices are mutually influential, too, as each person brings together a variety of practice domains.

Currently, a much broader range of people than ever before engage in photographic activity of all kinds, more often, under a greater variety of conditions, making more, and more varied, images. Participants include not just photographers but subjects and viewers, technology designers and producers – anyone who engages in photo-related, photographic-technology-related

activities. Many people engage in photography within the framework of social media, which, I argue, torques photographic practices in that direction. Practices are primarily tacit, learned through seeing, doing and talking. Both the persistence of existing practices and the development of new ones require visible, public performance. The practices visible via photographs are of two kinds: photographic practices and the practices made visible *in* photographs. For example, Uimonen's (this volume) Tanzanian mother's photos make visible how she performs her culture's practices of motherhood. Morcate and Pardo (this volume) describe the making and sharing of images of illness, grief and death. These images don't only represent a change in the contemporary photographic practices in Western countries, where such images have been seen as inappropriate. They also depict changes in the practices around illness and death: 'there is no longer a necessity to relegate grief to an intimate environment' (Morcate & Pardo). Put these together and, I argue, we are seeing important implications for photography practice from networked everyday photography. First, it's likely that ongoing photographic practices will have more and more in common with social media and potentially less with traditional photography. Second, I argue that networked photography contributes to more rapid innovation and dissemination in practices in many domains. While this may result in more uniformity across cultures, these chapters give me hope that we will see more diversity in different places and conditions.

Another notable development is that photos have become, paradoxically, both more personal and more collective. Most of the chapters in this volume address the blurring of the boundary between public and private images and the increasing publicness of images. Many chapters describe the immediate, shared uses of networked images and video, up to and including live streaming – what Favero (in this volume) calls 'shared simultaneous experience'. As images (and video) become more collective and immediate, viewers experience more points of view, not just locational but experiential and subjective. We get the visceral experience of seeing (and, with video, hearing) events unfolding in front of our own eyes, even in real time. We see sights usually inaccessible to outsiders, such as from civilians and soldiers in war zones. People publish images and videos online that the mass media cannot or will not publish (Posetti 2014). Furthermore, many shared, online images are indexed or tagged, and thus more easily retrieved than one's own. Why trawl through an unorganized, unlabelled personal collection of digital files when similar images are readily findable online? In fact, some people ask, why make or keep an image when other people, better situated, possibly more skilled, with better equipment, have made better images available?

However, image-capture devices and storage technologies have also become more personal, making some images inaccessible to anyone but the owner. The household camera has been replaced by personal cameras, smartphones and tablets, and even wearable devices. The images in the photo album or box of prints or slides have been dispersed to individuals'

cameras, smartphones, hard drives, social media sites and Cloud storage. The massive increase in image-making, not to mention developments like life-logging (Favero, this volume), threaten to bury us in unorganized, undifferentiated images, which are difficult to retrieve (if anyone even cares to try) both logistically and legally after the passing of the owner.

A major traditional use for photos that is threatened, I argue, is personal and collective memory. Blustein (2008), a moral philosopher, goes so far as to say that personal and collective memory is a moral imperative: to know who we are requires that we know where we came from; and we must take responsibility for our past. Bowker (2005) describes how durable memory technologies, objects and practices 'permit both the creation of a continuous, useful past and the transmission *sub rosa* of information, stories and practices from our wild, discontinuous, ever-changing past' (p. 9). He terms 'inaugural acts' (p. 10) disruptions such as the invention of printing that create disjunctions in archives and practices.

The Achilles heel of digital files is durability and retrievability (Van House & Churchill 2008), exacerbated by an emphasis on immediacy and transiency in image making and use. Internet pioneer Vint Cerf has warned of a coming 'digital dark age' (Pultarova 2015) as archiving technologies continue to change and older files become inaccessible. I fear that, as inaugural acts, digital technologies and social media have disrupted our memory regimes and the integrity of personal and collective archives and threaten images' traditional role as memory objects.

My point is not that images are no longer useful or used for memory, collective and individual. Indeed, as more images are made and shared more widely, the visual record is expanding. However, what we can 'remember' via the visual record depends on the images made, archived, indexed and migrated as technologies change. Much is gained, but what is lost with technologies and practices that emphasize immediacy over durability? For example, the Holocaust Museum in Washington, DC, solicits private photos of life in Europe between the Wars to document an important era in European and Jewish history. A box of glass negatives found in a garage proved to be a valuable record of African-Americans' lives in the US Midwest in the 1910s and 1920s. What happens when such images are lost because they are digital files that disappear when they aren't migrated to new storage media, or a Cloud storage service goes out of business? Or are never made because people were relying on others' images?

Finally, I want to comment on networked photography as performance: performance of photographic practices (e.g. Becker, this volume), performance in front of the camera (e.g. Andersen) and, in a different use of the term, performance of self or group, actual or idealized (Blanco; Uimonen; Mota; Villi; Andersen; Thielman), or space (Andersen; Pink; Thielmann; Villi). Even before social media, much of photography was performance for an actual or imagined audience, not just by the subjects

but the photographer as well. Photography is never mere representation or transcription: it is performance and enactment by all involved.

Self-aware performance in front of a camera has always made a photo a part of the event being photographed. Crang (1997) argued that travel photography (his particular interest)

... sacrifice[s] the immediacy of experience and orientat[es] activities to (future, distant) viewers. Events are framed for the future perfect, to have been ... *We cannot look on the photo as simply recording the event when it is part of that event's very nature.*

(pp. 365–366, emphasis added)

A further hazard of networking is that people may find themselves represented by others, pictured online without their agreement or even knowledge. Becker (this volume) reminds us of the power relations built into networked images. For example, adolescents may agonize every morning over what to wear, knowing that their photos may be taken and posted online. More troubling are the instances where, for example, sexual assaults are filmed and posted online.

Another kind of performance associated with photographs and photography is that of the photographic objects and technologies themselves. The current ontological turn in Science and Technology Studies (STS) means that any entity is multiple. No object or technology is pre-given; it may be enacted in different versions by different people under different circumstances, practices and social and material relations (Law & Lien 2013; Mol 2002, 2013; Woolgar & Lezaun 2013; Van House 2015). These chapters demonstrate many of the different versions, in ontological terms, of photographs and photographic technologies that are enacted currently.

Finally, and perhaps most importantly, there is the performance or enactment of identity. Poletti and Rak say of social media that, from the perspective of auto/biography studies, the self is 'an effect of representation ... rather than one's identity being expressed through online practices ... identity is in fact a product of the writing or composing process' (2014, p. 6). People don't *represent* but *enact* themselves in the process of picturing (literally) themselves online. Elsewhere (Van House 2011) I have demonstrated how the design of social media sites influences the kinds of selves (and places) that can be enacted, are encouraged and supported via the designs and practices of current and developing technologies? What does this mean for who we are and will become? These questions are entangled with the issue of memory: what are the stories that we tell ourselves about our past? That can be told with current technologies? How do these interact with current enactment of identity?

Research theory and methods

This book is also concerned with theory and research methods associated with images, materiality and visibility. Theory, research questions, assumptions and methods are all intertwined.

Research methods

As these chapters demonstrate, practice-based research is largely empirical and qualitative. The downside of qualitative research is that it can only effectively engage with small numbers of people in highly specific situations. Generalization is contrary to the theoretical stance that practices are highly contingent. For example, Uimonen (this volume) shows that Tanzanians' resistance to being photographed by strangers, especially white people, has roots in their country's colonial history and current processes of cultural commodification tied to social inequality.

New empirical research methods take advantage of the large collections of images now available online. Early researchers (e.g. Chalfen 2004; Rose 2003) went into households to see their photos. Now, online services like Flickr and Instagram provide easy access to millions of images, often with metadata and linkages to people and other images. Big data analysis methods are being brought to bear on these collections. Computer vision research, facilitated by these huge corpuses, can recognize elements within photos and even identify faces (a technology that is now filtering down to consumers). Researchers are beginning to use 'big data' statistical analyses to try to understand the content as well as the activity around these images. For example, from a selection of 23 million Instagram images from three million users, Bakhshi, Shamma and Gilbert (2014) concluded that people are more interested in (i.e., they 'like' and comment on) images with faces than those without. These enormous collections of linked images and metadata offer ways to study not just photographs, but the processes of networked photography over time: for example, the kinds of images made, the cameras used and patterns of viewing, linkages, sharing and tags.

From a practice perspective, a major failing of big data research is that it removes images from social contexts and practices. It ignores the situated nature of images. Although such analyses can be useful, it is important to remember what they can and cannot tell us. The inferences computer vision analysis can make are limited. These are only images that owners have put online; access depends on owners' privacy choices and institutional policies. User-supplied metadata such as tags are highly variable both in content and reliability. Many of these corpuses are proprietary. Researchers working for service providers may be the only ones with access to certain collections. Ethical, policy and legal concerns further limit access. We will undoubtedly see a cascade of research on these large collections, but need to understand its limitations.

Theory

The chapters in this volume show influence from a variety of areas of theory, including media studies, geography and communications studies. The editors of a recent *Handbook of Visual Culture* (Sandywell & Heywood 2012) make a strong case for the need for inter- and even transdisciplinary research. They urge their own research community to:

... conceptualize 'the visual' less as a discretely bounded domain of study – traditionally occupied by art history, media studies, film studies, and the like – and more as an underlying problematic that informs the disciplinary concerns and research programmes of all the arts, sciences and philosophy. As an intersecting cluster of themes and problems – what we might call a transverse problematic – visual culture lends itself to a multiplicity of theoretical interventions and programmatic inquiries.

(p. 12)

Their advice applies to the research represented in this book as well: to consider what fields like art history, media studies, visual studies and others can offer.

In that same volume, Rose (2012), a cultural geographer who has done extensive empirical research on photographic practices (e.g. Rose 2010; also this volume), argues that the social sciences have much to offer to the field of visual culture. She notes that the discipline of visual culture focuses heavily on the meaning of images derived from the formal qualities of the visual object without discussing audiences or place of production or viewing, and without reflecting on the processes and conditions under which the critic produces claims about meaning. She argues that the field of visual culture can benefit from both theoretical and empirical attention to practices, that is, the kind of research represented by these chapters.

Finally, I want to highlight two noteworthy omissions in current research. First, surprisingly little research addresses high quality, stand-alone, non-networked¹ cameras, except to predict (or celebrate) the death of the point-and-shoot camera. Mirrorless cameras and dSLRs make high quality images and sophisticated functionality readily available to the everyday photographer. Powerful, readily accessible software, from smartphone apps to Photoshop, enable sophisticated image processing. In my experience, while professional photographers and photojournalists are anxious about the future of their professions, many non-professionals are engaged in serious aesthetic and documentary photography, visual narrative and other such activities, previously not part of the everyday photographer's experience. Yet, most of the research on non-professional photography is concerned with more quotidian activity.

Second, a topic currently under-theorized in research on everyday photography is video, especially networked video made by everyday photographers

and videographers. Short videos made by mobile devices – we might call them video postcards or video snapshots – are an increasing proportion of networked visual media. In late 2015, Facebook began supporting seven-second videos as profile pictures. However, our understanding of short, networked video is, in some ways, like that of early mobile photography. In this volume, Andersen examines in depth the political uses of video and assumptions about its truthfulness, and Favero discusses the GoPro video device as a window on to other senses. There is much more to be learned about simple consumer-produced video.

As I said at the beginning, innovations and discontinuities help to make visible what has been taken for granted. As video becomes more a part of the ecology of everyday photography, it highlights some of the questions of material mediations raised in this volume. It makes visible some of the interactions among photographic and other kinds of practices, technologies and infrastructure. As infrastructural elements like Facebook and GoPro better support everyday video-making and viewing, uses and practices will emerge. Finally, video, especially live-streaming, raises new issues of ethics and values in design. Video can be more immediate, both temporally and emotionally. Terrorists, for example, display videos of executions online. Just because we can live-stream, should we? Who? When? And when not? Should we watch? (Malik 2015).

Conclusions

Finally, I ask two larger questions about photographic practices and images. The first is about how these developments affect people's lives. The second is about images themselves.

The phenomenological anthropologist Timothy Ingold describes life as wayfaring. People, he says, are in continual movement through the world.

Along such paths, lives are lived, skills developed, observations made and understandings grown.... To be, I would now say, is not to be in place but to be along paths. The path, and not the place, is the primary condition of being, or rather becoming.... My contention is that way-faring is the fundamental mode by which living beings inhabit the earth. Every such a being has, accordingly, to be imagined as the line of its own movement or – more radically – as a bundle of lines.

(2011, pp. 12–13)

The intersections of these paths of movements he calls meshwork. We all live in a meshwork of our own lines of movement and that of others, human and non-human. Building on Ingold, Pink (this volume) describes people as digital wayfarers, weaving through digital/virtual and material/physical worlds. Images are in movement, too:

... [Images] are on encounters with things ... [Images] are produced by moving through and not over or on environments, and they are not stopping points so much as outcomes of and in movement. This means making a shift in thinking from the commonsense idea that a photograph represents a static moment and thus is a photograph of a place, to the idea that it is in place and in movement.

(Pink 2011, p. 9, cited by Pink in this volume)

Photography, in all its versions, is entangled with people's wayfaring through the meshwork that is the world of living and technology. Ingold says that his field, anthropology, 'is the study of human becomings, as they unfold within the weave of the world' (2011, p. 9). So, too, the study of everyday photography is, in the end, like all social science, the study of human becomings, and how photographic technologies and practices are also in movement (Van House 2015), enmeshed with who we are and are becoming, individually and collectively.

Finally, the question that much of the discussion of photography circles around: why are *images* so important? Why are people so enamoured of photography? Why have new photographic technologies and practices been embraced so enthusiastically even while old ones have remained beloved? David Freedberg (1989, p. xxi) has argued that art history has erred in ignoring the recurring 'unrefined, basic, preintellectual, raw' ways in which people throughout history and cultures have responded to images of all kinds. We should consider

... the effectiveness, efficacy and vitality of images themselves; not only what beholders do, but also what images appear to do; not only what people do as a result of their relationship with imaged form, but also what they expect imaged form to achieve, and why they have such expectations at all.

(Ibid., p. xxii)

Media theorist W. J. T. Mitchell shares Freedberg's puzzlement.

Why is it that people have such strange attitudes toward images, objects, and media? Why do they behave as if pictures were alive, as if works of art had minds of their own, as if images had a power to influence human beings, demanding things from us, persuading, seducing, and leading us astray? ... How is it, in other words, that people are able to maintain a 'double consciousness' toward images, pictures, and representations in a variety of media, vacillating between magical beliefs and skeptical doubts, naive animism and hardheaded materialism, mystical and critical attitudes?

(2005, p. 7)

Empirical research on the large, diverse and ever-growing domain of everyday photographs – malleable and slippery as they are – and photographic practices and technologies, and the ways that they are entangled in the meshwork of people's lives, may help us further along towards finding answers to these questions.

Note

- 1 Many of the newest dSLRs are now Wi-Fi capable, but they differ significantly, as networked devices, from the ever-present camera phone.

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